Exercises of category theory

Filippo Rossi

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This document comprises a collection of exercises, accompanied by their respective solutions, pertaining to the course "Introduction to Categories" delivered by Professor Antoine Ducros at the Sorbonne University in Paris during the academic year 2025-2026. The exercise numbers match those in the course notes distributed by the professor during that academic year.

I compiled this document during my studies, and it is possible that it may contain errors. I would be grateful if you could notify me of any inaccuracies you may find.

Contents

1 Category, functors, morphisms of functor

Exercise 1 (Exercise 1.1.11)**.** Let $n \in \mathbb{N}_{\geq 1}$ and let B be the unit ball in $\mathbb{R}^n$. Show that the class of the inclusion $i : \{0\} \to B$ is an isomorphism of **hTop**

Solution 1. Let

$$\begin{array}{rccc} c_0 : & B & \to & \{0\} \\ & x & \mapsto & 0 \end{array}$$

the constant map. Then we have that $i \circ c_0 = \mathrm{Id}_{\{0\}}$. Moreover we have that $c_0 \circ i \approx \mathrm{Id}_B$, indeed we can built an homotopy as follows:

$$\begin{array}{rccc} \phi : & [0,1] \times B & \to & B \\ & (t, x) & \mapsto & tx \end{array}$$

Now $\phi(0, x) = c_0(x)$ and $\phi(1, x) = \mathrm{Id}_B(x)$. Thus, in **hTop**, we have that $\overline{c_0 \circ i} = \overline{\mathrm{Id}_B}$. Then i is an isomorphism.

Exercise 2 (Exercise 1.1.12)**.** Let $\mathcal{C}$ be a category and let $f : X \to Y$ be an arrow of $\mathcal{C}$. We say that f is a *monomorphism* if for every object Z of $\mathcal{C}$ the map $g \mapsto f \circ g$ from $\mathrm{Hom}_{\mathcal{C}}(Z, Y)$ to $\mathrm{Hom}_{\mathcal{C}}(Z, X)$ is injective; we say that f is an *epimorphism* if for every object Z of $\mathcal{C}$ the map $g \mapsto g \circ f$ from $\mathrm{Hom}_{\mathcal{C}}(Y, Z)$ to $\mathrm{Hom}_{\mathcal{C}}(X, Z)$ is injective.

1. Show that an isomorphism is both an epimorphism and a monomorphism.
2. In the category **Set** show that monomorphisms are injections, and that epimorphisms are surjections.
3. Show that monomorphisms of **Ab**, **Grp** and **Ring** are injections.
4. Show that epimorphisms of **Ab** and **Grp** are surjections (the case of **Grp** is delicate).
5. Give an example of an epimorphism of **Ring** that is not a surjection.

Solution 2.

1. Let $f : X \to Y$ be an isomorphism. Let's say that the following diagram

$$\begin{tikzcd} X \arrow[r, "f"] & Y \\ Z \arrow[u, "g"] & Z \arrow[ul, "h"] \end{tikzcd}$$

toggles. Then we have that $f \circ g = f \circ h$. Then have $g = f^{-1} \circ f \circ g = f^{-1} \circ f \circ h = h$. Then f is a monomorphis. Let's now say that the following diagram

$$\begin{tikzcd} Y \arrow[r, "g"] & Z \\ X \arrow[u, "f"] \arrow[r, "f"] & Y \arrow[u, "h"'] \end{tikzcd}$$

toggles. Then we have that $g \circ f = h \circ f$. Thus $g = g \circ f \circ f^{-1} = h \circ f \circ f^{-1} = h$. Then f is an epimorphism.

2. Let's deal with the monomorphisms.
$(\implies)$: Let $f : X \to Y$ be a monomorphism, and let $a, b \in X$ such that $f(a) = f(b)$. Let's define the following maps

$$\begin{array}{rccc} g_a : & \{\cdot\} & \to & X \\ & \cdot & \mapsto & a \end{array}$$

$$\begin{array}{rccc} g_b : & \{\cdot\} & \to & X \\ & \cdot & \mapsto & b \end{array}$$

Since f is a monomorphism and since we have that $f(a) = f \circ g_a(\cdot) = f \circ g_b(\cdot) = f(b)$ we have that $g_a = g_b$. Thus we have that $a = b$, thus f is injective.
$(\Longleftarrow)$: Let $f : X \to Y$ an injective map and let the following diagram

$$\begin{tikzcd} X \arrow[r, "f"] & Y \\ Z \arrow[u, "g"] & Z \arrow[ul, "h"] \end{tikzcd}$$

toggles, i.e. $f \circ g = f \circ h$. Let now $x \in Z$, then we have $f(g(z)) = f \circ g(z) = f \circ h(z) = f(h(z))$. Since f is injective, we have that $g(z) = h(z)$ for nay $z \in Z$. Then f is a monomorphism.
Now let's deal with the epimorphism.
$(\Longrightarrow)$: Let $f : X \to Y$ an epimorphism and let's define the following maps

$$\begin{array}{rccc} c_1 : & Y & \to & \{0,1\} \\ & y & \mapsto & 1 \end{array}$$

$$\begin{array}{rccc} \chi_{\text{Im}f} : & Y & \to & \{0,1\} \\ & y & \mapsto & \begin{cases} 1, \text{ if } y \in \text{Im}f \\ 0, \text{ if } y \notin \text{Im}f \end{cases} \end{array}$$

Now, since f is a epimorphism and since we have that $c_1 \circ f = \chi_{\text{Im}f} \circ f$ we have that $c_1 = \chi_{\text{Im}f}$. Then for any $y \in Y$, $1 = c_1(y) = \chi_{\text{Im}f}(y)$, thus $y \in \text{Im}f$. Thus f is surjective.
$(\Longleftarrow)$: Let $f : X \to Y$ a surjective map and let the following diagram

$$\begin{tikzcd} Y \arrow[r, "g"] & Z \\ X \arrow[u, "f"] \arrow[r, "f"] & Y \arrow[u, "h"'] \end{tikzcd}$$

toggles. Let now $y \in Y$, since f is surjective there exists $x \in X$ such that $f(x) = y$. Thus, since the diagram toggles, we have that $g(y) = g \circ f(x) = h \circ f(x) = h(y)$. Thus $h = g$ and we have the claim.

3. $(\Longrightarrow)$: Let $f : G \to H$ a monomorphism, let's show that it is injective. Let $g \in G$ such that $f(g) = e_H$. Let's define the following maps

$$\begin{array}{rccc} \phi : & \mathbb{Z} & \to & G \\ & n & \mapsto & g^n \end{array}$$

$$\begin{array}{rccc} \psi : & \mathbb{Z} & \to & G \\ & n & \mapsto & e_G \end{array}$$

Let $m \in \mathbb{Z}$, then $f \circ \phi(m) = f(g^m) = f(g)^m = e_H = f(e_G) = f \circ \psi(m)$. Thus $\phi = \psi$, then $g = \phi(1) = \psi(1) = e_G$. Then f is injective.

($\Longleftarrow$) : Let $f : X \to Y$ an injective map and let the following diagram

$$\begin{tikzcd} X \arrow[r, "f"] & Y \\ Z \arrow[u, "g"] & Z \arrow[ul, "h"] \end{tikzcd}$$

toggles, i.e. $f \circ g = f \circ h$. Let now $x \in Z$, then we have $f(g(z)) = f \circ g(z) = f \circ h(z) = f(h(z))$. Since f is injective, we have that $g(z) = h(z)$ for nay $z \in Z$. Then f is a monomorphism.

4. $f : X \to Y$ is an epimorphism if for any morphism $h, h' : Y \to H$ such that the diagram

$$\begin{tikzcd} Y \arrow[r, "h"] & H \\ X \arrow[u, "f"] \arrow[r, "f"] & Y \arrow[u, "h'"'] \end{tikzcd}$$

toggles we have that $h = h'$.

Ab: Let's f be an epimorphisms, fix $H = Y/f(X)$ and define $h = 0$ and $h' = \pi$ the projection on $f(X)$. Let now $x \in X$ be an element, then

$$h(f(x)) = 0 = \pi(f(x))$$

thus we have $0 = \pi$, that is $f(X) = Y$. Then f is surjective[1].
Conversely, let's suppose that f is surjective. Now if $y \in Y$ we have that $\exists x \in X$ such that $f(x) = y$ and then

$$h(y) = h \circ f(x) = h' \circ f(x) = h'(y)$$

Hence $h = h'$ and we have an epimorphism.

Grp : First of all, let's consider $\Omega := Y/f(X)$ the sets of right cosets of $f(X)$ in Y. Let ∞ be a symbol such that $\infty \notin \Omega$. Now let's define $\Omega^* := \Omega \cup \{\infty\}$ and define $H := \mathrm{Sym}(\Omega^*)$[2]. Now let's define the morphisms h and h':

- Let's define

$$\begin{aligned} h \ : Y \ &\to \qquad H \\ y \ &\mapsto \begin{pmatrix} \Omega^* & \to & \Omega^* \\ f(X)a & \mapsto & f(X)(ay) \\ \infty & \mapsto & \infty \end{pmatrix} \end{aligned}$$

that is a group homomrophism.

[1]We used that the grops are abelian to state that $f(X)$ is a normal subgroup of Y in order to take the quotient of Y.

[2]$\mathrm{Sym}(\Omega^*)$ is the group permutation of Ω^*

- Now let $\sigma* \in H$ the permutation such that swaps only $f(X)$ with ∞ and does not do anything on the other element. And we can define

$$\begin{array}{rrcc} h' & : Y & \to & H \\ & y & \mapsto & \sigma^{*-1} \circ h(y) \circ \sigma^* \end{array}$$

since h' is obtained from h by conjugation by σ , it is also a group homomorphism.

Now let's prove that $h \circ f = h' \circ f$. Let's $x \in X$, thus we have that

$$h(f(x))(f(X)) = f(X)f(x) = f(X)$$

hence $f(X)$ is fixed by $h(f(x))$. Since σ^* only swaps $f(X)$ with ∞ we have that $h(f(x))$ and σ^* have disjoint supports; thus they commutes, hence

$$h'(f(x)) = \sigma^{*-1} \circ h(f(x)) \circ \sigma^* = \sigma^{*-1} \circ \sigma^* \circ h(f(x)) = h(f(x))$$

Since f is an epimorphism we have that $h = h'$, hence for any $y \in Y$

$$h(y) = \sigma^{*-1} \circ h(y) \circ \sigma^*$$

that is

$$\sigma^* \circ h(y) = h(y) \circ \sigma^*$$

Hence we have proved that two permutations commute, and this is true if and only if their support is disjoint (since they are not the inverse element of the each other). Since σ^* moves $f(X)$ we have that $h(y)$ fix $f(X)$, thus

$$f(X) = h(y)(f(X)) = f(X)y$$

This means that $y \in f(X)$ for any $y \in Y$. Hence $f(X) = Y$ ande we have the surjectivty of f.
Conversely, let's suppose that f is surjective. Now if $y \in Y$ we have that $\exists x \in X$ such that $f(x) = y$ and then

$$h(y) = h \circ f(x) = h' \circ f(x) = h'(y)$$

Hence $h = h'$ and we have an epimorphism (this proof is equal to the previous one).

5. Let us consider the inclusion $i : \mathbb{Z} \to \mathbb{Q}$. This is clearly non surjective, let's show now that is an epimorphism. Let us say that the following diagram

$$\begin{tikzcd} \mathbb{Q} \arrow[r, "g"] & X \\ \mathbb{Z} \arrow[u, "i"] \arrow[r, "i"] & \mathbb{Q} \arrow[u, "h"'] \end{tikzcd}$$

toggles for some X ring and for some $g, h : \mathbb{Q} \to X$ homorphism of rings. Let now $\frac{a}{b} \in \mathbb{Q}$. Now we have that $h(b)h(\frac{a}{b}) = h(b\frac{a}{b}) = h(a) = h \circ i(a) = g \circ i(a) = g(a) = g(b\frac{a}{b}) = g(b)g(\frac{a}{b})$. Now since the diagram toggles we have that $g(b) = h(b)$. Thus we have $h(b)h(\frac{a}{b}) = h(b)g(\frac{a}{b})$. Moreover, we have that $h(b) \cdot h(\frac{1}{b}) = h(\frac{b}{b}) = 1_X$, then $h(b)$ is invertible. Thus we have $h(\frac{a}{b}) = g(\frac{a}{b})$. Then i is an epimorphism.

Exercise 3 (Exercise 1.2.8)**.** Show that any fully faithful functor $F : \mathcal{C} \to \mathcal{D}$ is conservative.

Solution 3. Since F is fully faithful, we have that for any $X, Y \in \mathbf{Ob}(\mathcal{C})$ the map

$$\begin{array}{ccc} \mathbf{Hom}_{\mathcal{C}}(X,Y) & \to & \mathbf{Hom}_{\mathcal{D}}(F(X),F(Y)) \\ f & \mapsto & F(f) \end{array}$$

is bijective. We have to prove that F is conservative. Then let $f \in \mathbf{Hom}_{\mathcal{C}}(X,Y)$ and suppose that $F(f)$ is an isomorphism. Then there exists $g \in \mathbf{Hom}_{\mathcal{C}}(F(Y),F(X))$ such that $F(f) \circ g = \mathrm{Id}_Y$ and such that $g \circ F(f) = \mathrm{Id}_{F(X)}$. Since F is a full functor, there exists $h \in \mathbf{Hom}_{\mathcal{C}}(Y,X)$ such that $F(h) = g$. Let's show that $h = f^{-1}$:

$h \circ f = \mathrm{Id}_{\mathrm{X}}$: We have that $F(h \circ f) = F(h) \circ F(f) = g \circ F(f) = \mathrm{Id}_{F(X)} = F(\mathrm{Id}_X)$. Since F is a faithful functor $h \circ f = \mathrm{Id}_X$.

$f \circ h = \mathrm{Id}_{\mathrm{Y}}$: We have that $F(f \circ h) = F(f) \circ F(h) = F(f) \circ g = \mathrm{Id}_{F(Y)} = F(\mathrm{Id}_Y)$. Since F is a faithful functor $f \circ h = \mathrm{Id}_Y$.

Thus we have the claim.

Exercise 4 (Exercise 1.2.9)**.** Let X, Y be two isomorphic object in $\mathcal{C}$. Show that $\mathrm{Aut}X \cong \mathrm{Aut}Y$ as groups.

Solution 4. Let $f \in X \to Y$ the isomorphism. Thus we can define

$$\begin{array}{rccc} \Psi : & \mathrm{Aut}X & \to & \mathrm{Aut}Y \\ & g & \mapsto & f \circ g \circ f^{-1} \end{array}$$

Let's show that Ψ is an homomorphism of groups: let $g, h \in \mathrm{Aut}X$, then $\Psi(g \circ h) = f \circ g \circ h \circ f^{-1} = f \circ g \circ f^{-1} \circ f \circ h \circ f^{-1} = \Psi(g) \circ \Psi(h)$. Moreover the map

$$\begin{array}{rccc} \Phi : & \mathrm{Aut}Y & \to & \mathrm{Aut}X \\ & g & \mapsto & f^{-1} \circ g \circ f \end{array}$$

is again a homomorphism of group and it's the inverse of Ψ:

$$\Psi \circ \Phi(g) = \Psi(f^{-1} \circ g \circ f) = f \circ f^{-1} \circ g \circ f \circ f^{-1} = g$$

$$\Phi \circ \Psi(g) = \Phi(f \circ g \circ f^{-1}) = f^{-1} \circ f \circ g \circ f^{-1} \circ f = g$$

Exercise 5 (Exercise 1.2.10)**.** Show that the quotient functor $Q : \mathbf{Gp} \to \mathbf{OutGp}$ is conservative.

Solution 5. Let's study this functor:

$$\text{for any } X \in \mathrm{Ob}(\mathbf{Gp}),\ Q(X) = X$$

$$\text{for any } f \in \mathrm{Hom}_{\mathbf{Gp}}(X, Y),\ Q(f) = \overline{f}$$

Let's now $f \in \mathrm{Hom}_{\mathbf{Gp}}(X, Y)$ such that $Q(f) = \overline{f}$ an isomorphism in $\mathrm{Hom}_{\mathbf{OutGp}}(Q(X), Q(Y))$. Then there exists $\overline{h} \in \mathrm{Hom}_{\mathbf{OutGp}}(Q(Y), Q(X))$ such that $\overline{h} \circ \overline{f} = \mathrm{Id}_{Q(X)}$ and $\overline{f} \circ \overline{h} = \mathrm{Id}_{Q(Y)}$. Thus, from the properties of the functor we have:

$$\begin{aligned} \overline{h \circ f} = Q(h \circ f) = Q(h) \circ Q(f) = \overline{h} \circ \overline{f} = \mathrm{Id}_{Q(X)} = Q(\mathrm{Id}_X) = \overline{\mathrm{Id}_X} \\ \overline{f \circ h} = Q(f \circ h) = Q(f) \circ Q(h) = \overline{f} \circ \overline{h} = \mathrm{Id}_{Q(Y)} = Q(\mathrm{Id}_Y) = \overline{\mathrm{Id}_Y} \end{aligned}$$

Then there exist $u_X \in \mathrm{Aut}(X)$ and $u_Y \in \mathrm{Aut}(Y)$ such that

$$\begin{aligned} h \circ f = u_X \circ \mathrm{Id}_X = u_X \quad &\Longrightarrow \quad (u_X \circ h) \circ f = \mathrm{Id}_X \\ f \circ h = u_Y \circ \mathrm{Id}_Y = u_Y \quad &\Longrightarrow \quad f \circ (h \circ u_Y) = \mathrm{Id}_Y \end{aligned}$$

From the above identity we get that f has a left-inverse and a right-inverse, thus is an isomorphism.

Exercise 6 (Exercise 1.2.11)**.** Let k be a field. Let n and m be two integers.

1. What are the isomorphisms between n and m in the category $\mathbf{V}_k$ defined in 1.1.7 [3] ?
2. Is the functor $F : \mathbf{V}_k \to k\text{-Vect}_{\text{fin}}$ [4] full?
3. Faithful?
4. Fully faithful?

Solution 6.

1. Let $n, m \in \mathbb{N}$. If $f \in \text{Hom}_{n,m}$ is an isomorphism, then f is a matrix in $M_{n,m}(k)$ and exists $g \in M_{m,n}(k)$ such that $f \circ g = \mathbb{I}_n$ and such that $g \circ f = \mathbb{I}_m$. We clearly have that $\text{rank}(f \circ g) = \text{rank}(\mathbb{I}_n) = n$, from the properties of the rank we have that $n = \text{rank}(f \circ g) \leq \min\{n, m\}$, then we have that $n \leq m$. Moreover, we also have that $\text{rank}(g \circ f) = \text{rank}(\mathbb{I}_m) = m$, then (as we have done above) we get $m \leq n$. Thus we have that $n = m$. Then, if we want f to be invertible, we have to require that f is a matrix such that $\det(f) \neq 0$. Since we also have that a matrix $M \in M_{n,n}(k)$ such that $\det(M) \neq$ is an isomorphism in $\mathbf{V}_K$ we have that
$$\{\text{isomorphism in } \mathbf{V}_K\} = \{M \in M_{n,n}(k) \mid \det(M) \neq 0\}$$
2. Let $f : k^n \to k^m$ a linear map of vector space, i.e. a morphism in $\text{Hom}_{k\text{-Vect}_{\text{fin}}}(F(n), F(m))$. Naming as $e_1, \ldots, e_n$ the vectors of the canonical base for k^n we can write the matrix M_f^E with respect to the base $\{e_1, \ldots, e_n\}$. Since $F(M_f^E) = f$, F is a full functor.
3. Let $F(A) = F(B)$ for $A, B \in \mathrm{M}_{m,n}(k)$. Then, for any $v \in k^n$, we have that
$$A \cdot v = B \cdot v$$
Choosing $v = e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$ we get $A^i = A \cdot v = B \cdot v = B^i$ for any $i = 0, \ldots n$. Thus, $A = B$. Then we have that the functor is faithful.
4. From (2) and (3) follow that it is fully faithful.

[3] The category is: $\text{Ob}(\mathbf{V}_k) = \mathbb{N}$ and for any $m, n \in \mathbb{N}$ we have $\text{Hom}_{\mathbf{V}_k}(n, m) = M_{n,m}(k)$.

[4] The functor $F : \mathbf{V}_k \to k\text{-}\mathbf{Vect}^{\text{fin}}$ sends each natural number n to the vector space k^n, and each matrix $A \in \mathrm{M}_{nm}(k) = \text{Hom}_{\mathbf{V}_k}(m, n)$ to the corresponding linear map $F(A) : k^m \to k^n$, $v \mapsto A \cdot v$.

Exercise 7 (Exercise 1.3.16)**.** We work in the category of sets. We denote by F (resp. G) the covariant (resp. contravariant) functor from Ens to Ens such that $F(X) = \{*\}$ for all X (resp. $G(X) = \{*\}$ for all X) where $\{*\}$ is a singleton chosen once and for all. Show that:

1. $h_{\varnothing}$ is isomorphic to F
2. $h^{\{*\}}$ is isomorphic to G.
3. $h^{\varnothing}$ is isomorphic to a subfunctor of G that will be described.
4. $h_{\{*\}}$ is isomorphic to $\mathrm{Id}_{\mathrm{Ens}}$.

Solution 7.

1. Let's recall that
$$h_{\emptyset}(A) = \{p_A\}$$
is the only empty function $p_A : \emptyset \to A$. Let's define:
$$\begin{array}{ll} u : h_{\emptyset} \to F & \\ & \begin{array}{rccc} u(A): & h_{\emptyset}(A) & \to & F(A) \\ & p_A & \mapsto & * \end{array} \\ v : F \to h_{\emptyset} & \\ & \begin{array}{rccc} v(A): & F(A) & \to & h_{\emptyset}(A) \\ & * & \mapsto & p_A \end{array} \end{array}$$
Let's consider $f : A \to B$ with $A, B \in \mathrm{Ob}(\mathbf{Ens})$, thus:
$$\begin{tikzcd} h_{\emptyset}(A) \arrow[r, "h_{\emptyset}(f)"] \arrow[d, bend left, "u(A)"] & h_{\emptyset}(B) \arrow[d, bend left, "u(B)"] \\ F(A) \arrow[r, "F(f)"'] \arrow[u, bend left, "v(A)"] & F(B) \arrow[u, bend left, "v(B)"] \end{tikzcd}$$
Clearly the diagram toggles since they are all singletons they are isomorphisms.

2. Let's recall that
$$h^{\{*\}}(A) = \{p_A\}$$
is the only constant function $p_A : A \to \{*\}$. Let's now define:
$$\begin{array}{ll} u : h^{*} \to G & \\ & \begin{array}{rccc} u(A): & h^{\{*\}}(A) & \to & G(A) \\ & p_A & \mapsto & * \end{array} \\ v : G \to h^{\{*\}} & \\ & \begin{array}{rccc} v(A): & G(A) & \to & h^{\{*\}}(A) \\ & * & \mapsto & p_A \end{array} \end{array}$$

Let's consider $f : A \to B$ with $A, B \in \mathrm{Ob}(\mathbf{Ens})$, thus:

$$\begin{tikzcd} h^{\{*\}}(B) \arrow[r, "h^{\{*\}}(f)"] \arrow[d, bend left, "u(B)"] & h^{\{*\}}(A) \arrow[d, bend left, "u(A)"] \\ G(B) \arrow[r, "G(f)"'] \arrow[u, bend left, "v(B)"] & G(A) \arrow[u, bend left, "v(A)"] \end{tikzcd}$$

Again, the diagram toggles since they are all singleton and they are isomorphisms.

3. Let's recall that

$$h^{\emptyset}(A) = \{p_A\}, \text{ if } A = \emptyset$$
$$h^{\emptyset}(A) = \emptyset, \text{ if } A \neq \emptyset$$

where $p_A : A \to \emptyset$ is the only empty function. Let's now define:

$$\begin{array}{rccc} G' : & \mathbf{Ens}^{\mathrm{Op}} & \to & \mathbf{Ens} \\ & A & \mapsto & \begin{cases} \{*\}, & \text{if } A = \emptyset \\ \emptyset, & \text{if } A \neq \emptyset \end{cases} \end{array}$$

If we say that the functions are sent in the restriction, we have that G' is a sub-functor for G. Moreover, we can define:

$$u : h^{\emptyset} \to G'$$

$$\begin{array}{rcccl} u(A) : & h^{\emptyset}(A) & \to & G'(A) & \text{if } A = \emptyset \\ & p_A & \mapsto & * & \\ u(A) := p_{h^{\emptyset}(A)} : & h^{\emptyset}(A) & \to & G'(A) & \text{if } A \neq \emptyset \end{array}$$

$$v : G' \to h^{\emptyset}$$

$$\begin{array}{rcccl} v(A) := p_{G'(A)} : & G'(A) & \to & h^{\emptyset}(A) & \text{if } A \neq \emptyset \\ v(A) : & G'(A) & \to & h^{\emptyset}(A) & \text{if } A = \emptyset \\ & * & \mapsto & p_A & \end{array}$$

Let's consider $f : A \to B$ with $A, B \in \mathrm{Ob}(\mathbf{Ens})$, thus:

$$\begin{tikzcd} h^{\emptyset}(B) \arrow[r, "h^{\emptyset}(f)"] \arrow[d, bend left, "u(B)"] & h^{\emptyset}(A) \arrow[d, bend left, "u(A)"] \\ G'(B) \arrow[r, "F(f)"'] \arrow[u, bend left, "v(B)"] & G'(A) \arrow[u, bend left, "v(A)"] \end{tikzcd}$$

Again, the diagram toggles since they are all singleton and they are isomorphisms.

4. Let's recall that

$$h_{\{*\}}(A) = \{f_x \mid x \in A\}$$

where

$$\begin{array}{rccc} f_x : & \{*\} & \to & A \\ & * & \mapsto & x \end{array}$$

Moreover, we can define:

$$
\begin{array}{lrcccl}
u : h_{\{*\}} \to \mathrm{Id}_{\mathbf{Ens}} & & & & & \\
& u(A): & h_{\{*\}}(A) & \to & \mathrm{Id}_{\mathbf{Ens}}(A) & \text{if } A \neq \emptyset \\
& & f_x & \mapsto & x & \\
& u(A) := p_{h_{\{*\}}(A)}: & h_{\{*\}}(A) & \to & \mathrm{Id}_{\mathbf{Ens}}(A) & \text{if } A = \emptyset \\
v : \mathrm{Id}_{\mathbf{Ens}}(A) \to h_{\{*\}} & & & & & \\
& v(A) := p_A: & \mathrm{Id}_{\mathbf{Ens}}(A) & \to & h_{\{*\}}(A) & \text{if } A = \emptyset \\
& v(A): & \mathrm{Id}_{\mathbf{Ens}}(A) & \to & h_{\{*\}} & \text{if } A \neq \emptyset \\
& & x & \mapsto & f_x &
\end{array}
$$

Let's consider $f : A \to B$ with $A, B \in \mathrm{Ob}(\mathbf{Ens})$, thus:

$$
\begin{tikzcd}
h_{\{*\}}(A) \arrow[r, "h^{\emptyset}(f)"] \arrow[d, bend left, "u(A)"] & h_{\{*\}}(B) \arrow[d, bend left, "u(B)"] \\
\mathrm{Id}_{\mathbf{Ens}}(A) \arrow[r, "F(f)"'] \arrow[u, bend left, "v(A)"] & \mathrm{Id}_{\mathbf{Ens}}(B) \arrow[u, bend left, "v(B)"]
\end{tikzcd}
$$

The diagram toggles and they are isomorphisms.

Exercise 8 (Execrise 1.3.17)**.** We work in the category of sets. Let F be the contravariant functor from Ens to itself given by the formulas

$$F(X) = \mathcal{P}(X) \text{ and } F(f) = (\mathcal{P}(Y) \to \mathcal{P}(X), Z \mapsto f^{-1}(Z))$$

for all $X \in$ Ens and all maps $f : X \to Y$. Construct an isomorphism $u : h^{\{0,1\}} \to F$. Let τ be the bijection $0 \mapsto 1, 1 \mapsto 0$ from $\{0,1\}$ to itself. Describe the automorphism $u \circ \tau_* \circ u^{-1}$ of F.

Solution 8. Let's define the morphisms:

$$\begin{array}{llccc}
u : h^{\{0,1\}} \to F & & & & \\
 & u(A): & h^{\{0,1\}}(A) & \to & \mathcal{P}(A) \\
 & & \varphi & \mapsto & \varphi^{-1}(\{1\}) \\
v : F \to h^{\{0,1\}} & & & & \\
 & v(A): & \mathcal{P}(B) & \to & h^{\{0,1\}}(B) \\
 & & L & \mapsto & \chi_L
\end{array}$$

Let's prove that it is a functor morphism, let $f : A \to B$ for $A, B \in \mathrm{Ob}(\mathbf{Ens})$:

$$\begin{tikzcd}
h^{\{0,1\}}(B) \arrow[r, "h^{\{0,1\}}(f)"] \arrow[d, bend left, "u(B)"] & h^{\{0,1\}}(A) \arrow[d, bend right, "u(B)"'] \\
\mathcal{P}(B) \arrow[r, "F(f)"'] \arrow[u, bend left, "v(B)"] & \mathcal{P}(A) \arrow[u, bend right, "v(B)"']
\end{tikzcd}$$

Let $\psi \in h^{\{0,1\}}(B)$ and $L \in \mathcal{P}(B)$, thus:

$$\begin{aligned}
u(B)(h^{\{0,1\}}(f)(\psi)) = u(B)(\psi \circ f) = (\psi \circ f)^{-1}(\{1\}) = f^{-1}(\psi^{-1}(\{1\})) = F(f)(\psi^{-1}) = F(f)(u(B)(\psi)) \\
v(B)(F(f)(L)) = v(B)(f^{-1}(L)) = \chi_{f^{-1}(L)} = \chi_L \circ f = h^{\{0,1\}}(f)(\chi_L) = h^{\{0,1\}}(f)(v(B)(L))
\end{aligned}$$

Thus they are morphisms of functors. Moreover:

$$\begin{gathered}
(u(B) \circ v(B))(L) = u(B)(\chi_L) = \chi_L^{-1}(\{1\}) = L \\
(v(B) \circ u(B))(\psi) = v(B)(\psi^{-1}\{1\}) = \chi_{\psi^{-1}\{1\}} = \psi
\end{gathered}$$

Then they are isomorphisms. Moreover, let $L \in \mathcal{P}(B)$, then

$$\begin{aligned}
(u(B) \circ \tau_*(B) \circ u^{-1}(B))(L) &= (u(B) \circ \tau_*(B) \circ v(B))(L) \\
&= (u(B) \circ \tau_*(B))(\chi_L) \\
&= u(B)(\tau \circ \chi_L) \\
&= (\tau \circ \chi_L)^{-1}(\{1\}) \\
&= \chi_L^{-1}(\tau^{-1}(\{1\})) \\
&= \chi_L^{-1}(\{0\}) = B \setminus L
\end{aligned}$$

Exercise 9 (Exercise 1.3.18)**.** We work in the category of groups. Let n be a positive or zero integer. Let F be the forgetful functor from Gp to Ens.

1. Construct an isomorphism from $h_{\mathbb{Z}/n\mathbb{Z}}$ to a subfunctor of F.

2. For which value of n is this subfunctor equal to F for all integers?

Solution 9.

1. Let's define the following functor:

$$\begin{array}{rccc} \mathcal{G}: & \mathbf{Gp} & \to & \mathbf{Set} \\ & G & \mapsto & \{x \in G \mid x^n = e_G\} \\ & (f: G \to H) & \mapsto & (f_{|\mathcal{G}(G)}: \mathcal{G}(G) \to \mathcal{G}(H)) \end{array}$$

Let's show that the functor is well define, i.e. let's show that $\mathcal{G}(f)$ maps in H if $f: G \to H$ is a group homomorphism. let now $x \in \mathcal{G}(G)$, then $x^n = e_G$. Thus

$$(\mathcal{G}(f)(x))^n = f(x^n) = f(e_G) = e_H$$

Since $e_H^n = e_H$ we have that the functor is well defined. Since $\mathcal{G}(G) \leq G$ for any group G, and $\mathcal{G}(f)$ is the restriction of f, we have that $\mathcal{G}$ is a sub-functor of the forgetful functor. Now let's show that the followings

$$U: \; h_{\mathbb{Z}/n\mathbb{Z}} \implies \mathcal{G}$$

$$\text{for any } A \in \mathrm{Ob}(\mathbf{Gp}), \quad \begin{array}{rccc} U(A): & h_{\mathbb{Z}/n\mathbb{Z}}(A) = \mathrm{Hom}_{\mathbf{Gp}}(\mathbb{Z}/n\mathbb{Z}, A) & \to & \mathcal{G}(A) \\ & \varphi & \mapsto & \varphi(1) \end{array}$$

$$V: \; \mathcal{G} \implies h_{\mathbb{Z}/n\mathbb{Z}}$$

$$\text{for any } A \in \mathrm{Ob}(\mathbf{Gp}), \quad \begin{array}{rccc} V(A): & \mathcal{G}(A) & \to & h_{\mathbb{Z}/n\mathbb{Z}}(A) = \mathrm{Hom}_{\mathbf{Gp}}(\mathbb{Z}/n\mathbb{Z}, A) \\ & x & \mapsto & (1 \mapsto x) \end{array}$$

are natural trasformations. Let $A, B \in \mathrm{Ob}(\mathbf{Gp})$ and $f: A \to B$ a group homomorphism, thus :

$$\begin{tikzcd} h_{\mathbb{Z}/n\mathbb{Z}}(A) \arrow[r, "h_{\mathbb{Z}/n\mathbb{Z}}(f)"] \arrow[d, bend left, "U(A)"] & h_{\mathbb{Z}/n\mathbb{Z}}(B) \arrow[d, bend left, "U(A)"] \\ \mathcal{G}(A) \arrow[u, bend left, "V(A)"] \arrow[r, "\mathcal{G}(f)"'] & \mathcal{G}(B) \arrow[u, bend left, "V(A)"] \end{tikzcd}$$

Let $\psi \in h_{\mathbb{Z}/n\mathbb{Z}}(A)$ and $x \in \mathcal{G}(A)$, then

$\mathcal{G}(f)(U(A)(\psi)) = \mathcal{G}(f)(\psi(1)) = f(\psi(1)) = (f \circ \psi)(1) = U(A)(f \circ \psi) = U(A)(h_{\mathbb{Z}/n\mathbb{Z}}(f)(\psi))$

$h_{\mathbb{Z}/n\mathbb{Z}}(f)(V(A)(x)) = h_{\mathbb{Z}/n\mathbb{Z}}(f)((1 \mapsto x)) = f \circ (1 \mapsto x) = (1 \mapsto f(x)) = V(A)(f(x)) = V(A)(\mathcal{G}(f)(x))$

Now we only have to show that U is the inverse of V. Let $\psi \in h_{\mathbb{Z}/n\mathbb{Z}}(A)$ and $x \in \mathcal{G}(A)$, then:

$$(V(A) \circ U(A))(\psi) = V(A)(\psi(1)) = (1 \mapsto \psi(1)) = \psi$$
$$(U(A) \circ V(A))(x) = U(A)((1 \mapsto x)) = x$$

then we have that $V \circ U = \mathrm{Id}_{h_{\mathbb{Z}/n\mathbb{Z}}}$ and $U \circ V = \mathrm{Id}_{\mathcal{G}}$.

2. Let $n > 0$, then we have that $\mathcal{G}(\mathbb{Z}) = \{0\}$, then does not exists an n such that the two functors are equal.

2 Equivalences of categories

Exercise 10 (Exercise 2.1.5)**.** We said above that the properties we are interested in (in a given category) are most often invariant under isomorphism. Can you give an example (in the category of sets, say) of a property that is *not* invariant under isomorphism?

Solution 10.

Exercise 11 (Exercise 2.1.6)**.** Construct an isomorphism between the category of $\mathbb{F}_p$-vector spaces and that of abelian groups G all of whose elements are of p-torsion.

Solution 11. Let's define the following functor:

$$\begin{array}{cccc} F: & \mathbb{F}_p-\mathbf{Vec} & \to & \mathbf{Ab}_p \\ & (A,+,\cdot) & \mapsto & (A,+) \\ & (f:A\to B) & \mapsto & (f:A\to B) \end{array}$$

This is well defined since if $x\in A$, then $\sum_{i=1}^{n} x = px = 0\cdot x = 0$. Now, let's define the following functor:

$$\begin{array}{cccc} G: & \mathbf{Ab}_p & \to & \mathbb{F}_p-\mathbf{Vec} \\ & (A,+) & \mapsto & (A,+,\cdot) \\ & (f:A\to B) & \mapsto & (f:A\to B) \end{array}$$

where

$$\overline{n}\cdot_i v := \sum_{i=1}^{n} v$$

The scalar product is well defined since if $\overline{n}=\overline{m}$ then $n=kp+m$, thus

$$\overline{n}\cdot_i v = \sum_{i=1}^{n} v = \sum_{i=1}^{kp+m} v = \sum_{i=1}^{kp} v + \sum_{i=1}^{m} v = \sum_{i=1}^{m} v = \overline{m}\cdot_i v$$

Now we also have:

$$G(F(A,+,\cdot)) = G(A,+) = (A,+,\cdot_i)$$

let's show by induction on n that $\overline{n}\cdot v = \overline{n}\cdot_i v$. For $n=1$ it is true, hence we have

$$\overline{n}\cdot v = \sum_{i=1}^{n} v = \sum_{i=1}^{n-1} v + v = \overline{(n-1)}\cdot_i v + v = \overline{n}\cdot_i v$$

Thus

$$(A,+,\cdot) = (A,+,\cdot_i)$$

Now we have

$$G(F(A,+)) = G(A,+,\cdot_i) = (A,+)$$

Then we have that F is an isomorphism.

Exercise 12 (Exercise 2.1.7)**.** Construct an isomorphism between the category of $\mathbb{Q}$-vector spaces and that of abelian groups G that are *uniquely divisible*, that is, such that for all $(g, n) \in G \times (\mathbb{N} \backslash \{0\})$ there exists a unique $h \in G$ such that $nh = g$ (sometimes we will write $h := h_{(g,n)}$).

Solution 12. Let's define the following functor

$$\begin{array}{cccc} F: & \mathbb{Q}-\mathbf{Vec} & \to & \mathbf{Ab}_{\text{ud}} \\ & (A,+,\cdot) & \mapsto & (A,+) \\ & (f : A \to B) & \mapsto & (f : A \to B) \end{array}$$

To show that the functor is well defined we have to prove that if A is a $\mathbb{Q}$ vector space, then the additive group A is a uniquely divisible group. Let $x \in A$ and $n \in \mathbb{N}_{>0}$, defining $h := \frac{1}{n} \cdot x$ we have that $n \cdot h = x$. Now we have to prove that such a h is unique. Say that $g \in A$ is another element such that $n \cdot g = x$, then we have $g = \frac{1}{n} \cdot x = h$. Then the functor is well-defined.
Let's now define the following functor:

$$\begin{array}{cccc} G: & \mathbf{Ab}_{\text{ud}} & \to & \mathbb{Q}-\mathbf{Vec} \\ & (A,+) & \mapsto & (A,+,\cdot_i) \\ & (f : A \to B) & \mapsto & (f : A \to B) \end{array}$$

where

$$\frac{a}{b} \cdot_i v := h_{(av,b)}$$

Let's prove that the functor is well defined, let $\frac{a}{b} = \frac{c}{d}$, then we have

$$bh_{(av,b)} = av \iff h_{(av,b)} = \frac{a}{b}v \iff h_{(av,b)} = \frac{d}{c}v$$

From the uniqueness of $h_{(av,b)}$ follows that $h_{(av,b)} = h_{(cv,d)}$. Thus is well defined. Now we have

$$G(F(A,+,\cdot)) = G(A,+) = (A,+,\cdot_i)$$

Let's prove that $(A,+,\cdot) = (A,+,\cdot_i)$,

$$\frac{a}{b} \cdot_i v = h_{(av,b)}$$

where $h_{(av,b)}$ is the only element such that $bh_{(av,b)} = av$. Since we also have $h := \frac{a}{b}v$ that satisfies the condition, we have $h = h_{(av,b)}$. Then

$$\frac{a}{b} \cdot_i v = h_{(av,b)} = h = \frac{a}{b} \cdot v$$

Since we also have that

$$F(G(A,+)) = F(A,+,\cdot_i) = (A,+)$$

we have that F is an isomorphism.

Exercise 13 (Exercise 2.2.10)**.** Let C and D be two categories and let F be a (covariant) functor from C to D. Show that the following assertions are equivalent:

1. there exists a strictly full subcategory D' of D such that F is the composite of a category equivalence from C to D' and the inclusion of D' in D (a subcategory D' of D is said to be strictly full if it is full and if every object of D isomorphic to an object of D' is an object of D');

2. F is fully faithful.

Solution 13.

($1 \implies 2$): We know that $F := i \circ e$, where $e : \mathcal{C} \to \mathcal{D}'$ is a category equivalence and $i : \mathcal{D}' \to \mathcal{D}$ is the inclusion. Thus we have that following diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, "e"] \arrow[dr, "F"'] & \mathcal{D}' \arrow[d, "i"] \\ & \mathcal{D} \end{tikzcd}$$

toggles. Once we have proved that both i and e are fully faithful, thus also F it's fully faithful and we have done. Since e is a category equivalence, we have that e is fully faithful and essentially surjective. Thus we only have to prove that i is fully faithful. Let $X, Y \in \mathrm{Ob}(\mathcal{D}')$ and let consider the map

$$\begin{array}{cccc} \phi : & \mathrm{Hom}_{\mathcal{D}'}(X,Y) & \to & \mathrm{Hom}_{\mathcal{D}}(X,Y) \\ & f & \mapsto & f \end{array}$$

Since $\mathcal{D}'$ is full we have that $\mathrm{Hom}_{\mathcal{D}'}(X,Y) = \mathrm{Hom}_{\mathcal{D}}(X,Y))$ and then ϕ is a bijection. Thus i is fully faithful and we ahve the claim.

($2 \implies 1$): Let's $\mathcal{D}'$ a subcategory for $\mathcal{D}$ as follows:

- the objects $\mathrm{Ob}(\mathcal{D}')$ are the objects Y of $\mathrm{Ob}(\mathcal{D})$ such that exists an X of $\mathrm{Ob}(\mathcal{C})$ such that $F(X) \cong Y$.
- given two objects X, Y of $\mathrm{Ob}(\mathcal{D}')$ we define the set $\mathrm{Hom}_{\mathcal{D}'}(X,Y) := \mathrm{Hom}_{\mathcal{D}}(X,Y)$.

We can see that $\mathcal{D}'$ is a strictly full subcategory of $\mathcal{D}$, indeed:

Strictly: let X an object of $\mathrm{Ob}(\mathcal{D})$ such that exists an object Y of $\mathrm{Ob}(\mathcal{D}')$ with $X \cong Y$. Since Y is an object of $\mathrm{Ob}(\mathcal{D}')$ there exists an object Z of $\mathrm{Ob}(\mathcal{C})$ such that $Y \cong F(Z)$. Thus we have $X \cong Y \cong F(Z)$, and we have that X is an object of $\mathrm{Ob}(\mathcal{D}')$;

Full: follows from the definition of the morphisms sets of $\mathcal{D}'$.

Now let's define the functor

$$\begin{array}{cccc} e : & \mathcal{C} & \to & \mathcal{D}' \\ & X & \mapsto & F(X) \\ & f : X \to Y & \mapsto & F(f) : F(X) \to F(Y) \end{array}$$

This is well defined since $F(X) \cong F(X)$, thus $F(X)$ is an object for $\mathrm{Ob}(\mathcal{D}')$. By construction we have that e is fully faithful (restriction of a fully faithful functor) and essentially surjective (we restricted the codomain to have this property). Thus we have that e is an equivalence of category. Clearly we also have that $F = i \circ e$.

3 Representable functors

Exercise 14 (Exercise 3.2.3)**.** Let k be a field and let C be the full subcategory of k-Alg formed by the extensions of k. Show that the forgetful functor from C to Set is not representable.

Solution 14. For the sake of the argument, let (X, ξ) with $X : \mathrm{Ob}(\mathcal{C})$ and $\xi \in F(X)$ be the representants for F, the forgetful functor. Thus we have that $h_X \overset{u_{F,\xi}}{\cong} F$, where the isomorphism ${}_{F,\xi} : h_X \to F$ is:

$$\text{for any } Y : \mathrm{Ob}(\mathcal{C}), \quad u_{F,\xi}(Y) : \quad \begin{array}{rcl} h_X(Y) = \mathrm{Hom}_{\mathcal{C}}(X, Y) & \to & F(Y) = Y \\ f & \mapsto & F(f)(\xi) = f(\xi) \end{array}$$

Now let's see the case $Y = k$. Since a morphism between fields is either injective or null, we have two possibilities:

1. If X does not admits immersion in k, then the only map that exists between X and k is the null map 0. Moreover, the map 0 does not fix the base field, thus we have

   $$\mathrm{Hom}_{\mathcal{C}}(X, k) = \emptyset$$

   Moreover we have
   $$\emptyset = \mathrm{Hom}_{\mathcal{C}}(X, k) = h_X(k) \cong F(k) = k$$
   that is a contradiction since $|k| \geq 2$.

2. If X admits an immersion f into k we have that $[X : k] \leq [k : k] = 1$, thus $X \cong k$ as $k-$algebras (from now we will consider $X = k$ under the identification through a fixed isomorphism). Thus we have

   $$\{\mathrm{Id}_k\} = \mathrm{Hom}_{\mathcal{C}}(k, k) = h_k(k) \cong F(k) = k$$

   that is again a contradiction since $|k| \geq 2$.

Exercise 15 (Exercise 3.2.4)**.** Let F be the functor from Ring to Set that assigns to a ring A the set of nilpotent elements of A. Show that F is not representable.

Solution 15. Suppose, for the sake of contradiction, that F is representable: there exists a ring $A \in \mathrm{Ob}(\mathbf{Ring})$ and an element $a \in F(A)$ (hence a is nilpotent) such that we have a natural isomorphism

$$u : h_A = \mathrm{Hom}_{\mathbf{Ring}}(A, -) \xrightarrow{\sim} F.$$

By the Yoneda lemma, u is determined by the element $a = u_A(\mathrm{id}_A) \in F(A)$, and for every ring B the bijection

$$u_B : \mathrm{Hom}_{\mathbf{Ring}}(A, B) \longrightarrow F(B), \qquad f \longmapsto F(f)(a)$$

holds. In particular, for every pair (B, b) with B a ring and $b \in F(B)$, there exists a *unique* ring homomorphism $f : A \to B$ such that $F(f)(a) = b$. Since $a \in F(A)$ is nilpotent, fix $k \in \mathbb{N}$ such that $a^k = 0$. For every $n \in \mathbb{N}$, set

$$B_n := \mathbb{Z}/p^{n+1}\mathbb{Z}, \qquad b_n := \overline{p} \in B_n.$$

Since $p^{n+1} = 0$ in B_n, the element b_n is nilpotent, i.e. $b_n \in F(B_n)$. By the universal property above, for each n there exists a unique ring homomorphism $f_n : A \to B_n$ such that $f_n(a) = \overline{p}$. Then

$$0 = f_n(a^k) = f_n(a)^k = \overline{p}^k \in \mathbb{Z}/p^{n+1}\mathbb{Z},$$

i.e. $p^{n+1} \mid p^k$, and therefore

$$n + 1 \leq k \qquad \text{for every } n \in \mathbb{N}.$$

But this is absurd: taking $n = k$ gives $k+1 \leq k$, which is false. Hence F is not representable.

Exercise 16 (Exercise 3.2.10)**.** Let G be a group and let E be a subset of G. Let F be the subfunctor of h^G that associates to a group H the set of morphisms $f\colon H \to G$ such that $f(H)$ is set-theoretically contained in E. Show that F is representable if and only if the set of subgroups of G contained in E admits a greatest element.

Solution 16. The functor h^G is the following functor:

$$\begin{array}{cccc} h^G : & \mathbf{Grp} & \to & \mathbf{Set} \\ & H & \mapsto & \mathrm{Hom}_{\mathbf{Grp}}(H,G) \\ & (f : H \to K) & \mapsto & \left(\begin{array}{cccc} f^* : & \mathrm{Hom}_{\mathbf{Grp}}(K,G) & \to & \mathrm{Hom}_{\mathbf{Grp}}(H,G) \\ & t & \mapsto & t \circ f \end{array} \right) \end{array}$$

Thus the subfunctor F wil be:

$$\begin{array}{cccc} F : & \mathbf{Grp} & \to & \mathbf{Set} \\ & H & \mapsto & \{t \in \mathrm{Hom}_{\mathbf{Grp}}(H,G) \mid t(H) \subset E\} \\ & (f : H \to K) & \mapsto & \left(\begin{array}{cccc} f^* : & F(K) & \to & F(H) \\ & t & \mapsto & t \circ f \end{array} \right) \end{array}$$

We have that F is representable if and only if exists $X \in \mathrm{Ob}(\mathbf{Grp})$ and $\xi \in F(X) = \{t \in \mathrm{Hom}_{\mathbf{Grp}}(X,G) \mid t(X) \subset E\}$ such that the morphism of functors

$$\begin{array}{cccc} v_{F,\xi} : & h^X & \to & F \\ v_{F,\xi}(Y) : & h^X(Y) = \mathrm{Hom}_{\mathbf{Grp}}(Y,X) & \to & F(Y) = \{t \in \mathrm{Hom}_{\mathbf{Grp}}(Y,G) \mid t(Y) \subset E\} \\ & (f : Y \to X) & \mapsto & F(f)(\xi) = f^*(\xi) \end{array}$$

is an isomorphism. Now let L the greatest subgroup of G such that $L \subset E$ (exists by hypothesis)and let $\xi = i_L$ be the inclusion of L in to G. Let's show that $v_{F,\xi}$ is a isomorphism in **Set** (that is a bijection):

Injectivity: Let $f, g : Y \to L$ two morphisms between groups such that $F(f)(i_L) = F(g)(i_L)$. Then we clearly have $f = g$.

Surjective: Let $t \in F(Y)$, thus we can change the codomain and obtain a map $t^* : Y \to L$ such that $F(t^*)(i_L) = t$

We have that F is representable [5] .

[5] We can see that the condition "the set of subgroups of G contained in E admits a greatest element" is necessary. Let's suppose that F is representable and let's set $G = \mathbb{Z}$ and $E = \{1\}$. Thus we would have that $h^X \cong F$ for some $X : \mathrm{Ob}(\mathbf{Grp})$. Thus we have:

$$X \cong \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}, X) = h^X(\mathbb{Z}) \cong F(\mathbb{Z}) = \{t \in \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z},\mathbb{Z}) \mid t(\mathbb{Z}) \subset \{1\}\} = \emptyset$$

This is a contradiction since X is a group and can not be an empty set.

Exercise 17 (Exercise 3.2.12)**.** Let Op be the contravariant functor from **Top** to **Set** such that $\mathrm{Op}(X)$ is for every X the set of open subsets of X, and such that for every continuous map $f\colon Y \to X$, the map $\mathrm{Op}(f)$ is given by the formula $U \mapsto f^{-1}(U)$. Show that Op is representable.

Solution 17. We have that

$$\begin{array}{rccc} \mathrm{Op}: & \mathbf{Top} & \to & \mathbf{Set} \\ & X & \mapsto & \{U \subset X \mid U \text{ is open in } X\} \\ & (f : X \to Y) & \mapsto & \begin{pmatrix} \mathrm{Op}(f): & \mathrm{Op}(Y) & \to & \mathrm{Op}(X) \\ & U & \mapsto & f^{-1}(U) \end{pmatrix} \end{array}$$

We have that Op is representable if and only if there exists a topological space X and an open set $\xi \in \mathrm{Op}(X)$ such that

$$\begin{array}{rccc} v_{\mathrm{Op},\xi}: & h^X & \to & \mathrm{Op} \\ v_{\mathrm{Op},\xi}(Y): & h^X(Y) = \mathrm{Hom}_{\mathbf{Top}}(Y, X) & \to & \mathrm{Op}(Y) \\ & (f : Y \to X) & \mapsto & \mathrm{Op}(f)(\xi) = f^{-1}(\xi) \end{array}$$

is an isomorphism of functor. Let's define $X = \{0, 1\}$ with the *Sierpínski* topology $\tau := \{\emptyset, \{1\}, X\}$ and $\xi = \{1\}$. Let's prove that it si bijective:

Injective: Let $f, g : Y \to X$ be two continuous function such that $f^{-1}(\{1\}) = \mathrm{Op}(f)(\{1\}) = \mathrm{Op}(g)(\{1\}) = g^{-1}(\{1\})$. Now let $y \in Y$, then $f(y) = 1$ if and only if $y \in f^{-1}(\{1\}) = g^{-1}(\{1\})$ if and only if $g(y) = 1$. Since we have also that $f(y) = 1$ if and only if $y \notin f^{-1}(\{1\}) = g^{-1}(\{1\})$ if and only if $g(y) = 0$ we have that $f = g$. Thus $v_{\mathrm{Op},\xi}(Y)$ is injective.

Surjective: Let $U \in \mathrm{Op}(Y)$, thus we can define the following map:

$$\begin{array}{rccc} f: & Y & \to & X \\ & y & \mapsto & \begin{cases} 0 & \text{if } y \notin U \\ 1 & \text{if } y \in U \end{cases} \end{array}$$

f is clearly continuous since $f^{-1}(\{1\}) = U$ is open. Moreover, $\mathrm{Op}(f)(\{1\}) = f^{-1}(\{1\}) = U$. Thus we have that $v_{\mathrm{Op},\xi}(Y)$ is surjective.

Thus the functor Op is representable.

4 Fiber products and amalgamated sums

Exercise 18 (Exercise 4.1.6)**.** Let C (resp. C_0, resp. C_p) be the full subcategory of **Ann** formed by fields (resp. fields of characteristic 0, resp. fields of characteristic p). Show that neither C, nor C_0, nor C_p admit cartesian products.

Solution 18. It suffices to show that does not exists the product between:

- $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2})$;
- $\mathbb{F}_2(x)$ and $\mathbb{F}_2(x)$;
- $\mathbb{Q}$ and $\mathbb{F}_2$.

Exercise 19 (Exercise 4.5.7)**.** Let C be a category in which finite products and coproducts exist. We would like to make sense of the following question: is the cartesian product distributive with respect to the coproduct?

1. Let (X, Y, Z) be a triple of objects of C. Show that the objects $X \times (Y \amalg Z)$ and $(X \times Y) \amalg (X \times Z)$ are related by a natural morphism (one of the two being the target and the other the source; it is up to you to decide which is which). We say that in C the cartesian product is distributive with respect to the coproduct if this natural morphism is an isomorphism for every such triple (X, Y, Z).

2. Define the dual notion of distributivity of the coproduct with respect to the cartesian product.

3. Examine the validity of these properties in the categories Ens and A-Mod.

4. Justify that an equivalence of categories preserves cartesian products and coproducts.

5. Show that Ens is not equivalent to $\mathsf{Ens}^{\mathrm{op}}$.

Solution 19.

1. Let's build a morphism from $(X \times Y) \coprod (X \times Z) \to X \times (Y \coprod Z)$. From the naturality of the cartesian coproduct, given two morphism $f : X \times Y \to X \times (Y \coprod Z)$ and $g : X \times Z \to X \times (Y \coprod Z)$ we get a unique morphism h such that the following diagram commutes:

$$\begin{tikzcd}
X \times (Y \coprod Z) & & \\
& (X \times Y) \coprod (X \times Z) \arrow[ul, "\exists!\, h"'] & X \times Z \arrow[l] \arrow[ull, bend right, "g"'] \\
& X \times Y \arrow[u] \arrow[uul, bend left, "f"] &
\end{tikzcd}$$

   Now let's build the morphism f (analogously we can construct the morphism g). From the naturality of the cartesian product, given two morphism $a : X \times Y \to X$ and $b : X \times Y \to Y \coprod Z$ we get a unique morphism $t : X \times Y \to X \times (Y \coprod Z)$ such that the following diagram commutes:

$$\begin{tikzcd}
X \times Y \arrow[dr, "\exists!\, t"] \arrow[drr, bend left, "j"] \arrow[ddr, bend right, "k"'] & & \\
& X \times (Y \coprod Z) \arrow[r] \arrow[d] & Y \coprod Z \\
& X &
\end{tikzcd}$$

Thus, we only have to build the morphisms k and j. Let's fix the following labels:

$$\begin{tikzcd} X \times Y \arrow[r, "q"] \arrow[d, "p"] & Y \\ X & \end{tikzcd}$$

$$\begin{tikzcd} Z \coprod Y & Y \arrow[l, "i_Y"'] \\ Z \arrow[u, "i_Z"] & \end{tikzcd}$$

Where the map are given from the definition of cartesian product and coproduct. Now we can define $k := p$ and $j := i_Y \circ q$. Now we have the morphism t, we can define $f := t$ (analogously we can get g) and we can get the morphism h.

2. The dual notion is

$$X \coprod (Y \times Z) \to (X \coprod Y) \times (X \coprod Z)$$

and can be proved in a similar way as before.

3. Let's deal with both cases:

Ens: We have to build a bijection between $(X \times Y) \sqcup (X \times Z)$ and $X \times (Y \sqcup Z)$:

$$\begin{array}{cccc} f: & (X \times Y) \sqcup (X \times Z) & \to & X \times (Y \sqcup Z) \\ & (x, h)_i & \mapsto & (x, h_i) \end{array}$$

where i is the tag (or label) indicating which summand of the disjoint union the element comes from. We clearly have that the map

$$\begin{array}{cccc} g: & X \times (Y \sqcup Z) & \to & (X \times Y) \sqcup (X \times Z) \\ & (x, h_i) & \mapsto & (x, h)_i \end{array}$$

is its inverse. Thus they are isomorphic.
The second property is $X \sqcup (Y \times Z) \cong (X \sqcup Y) \times (X \sqcup Z)$. Let's fix $X = \{1, 2\}$, $Y = \{3\}$ and $Y = \{4\}$. Thus we have that $X \sqcup (Y \times Z)$ has 3 elements, but the sets $X \sqcup Y$ and $X \sqcup Y$ have 3 elements; hence $(X \sqcup Y) \times (X \sqcup Y)$ has 9 elements. Thus they are not isomorphic.

$A -$ **Mod**: In this category, we have that:

$$X \prod Y = X \times Y$$
$$X \coprod Y = X \times Y$$

Thus it is clear that $(X \times Y) \times (X \times Z) \cong (X \times Y) \coprod (X \times Z)$ and $X \times (Y \coprod Z) \cong X \times (Y \times Z)$ can not be isomorphic. Indeed in the case $A = \mathbb{Z}$, $X = \mathbb{Z}/2\mathbb{Z}$, $Y = Z = \{0\}$ we have:

$$|(X \times Y) \times (X \times Z)| = 4$$
$$|X \times (Y \times Z)| = 2$$

Since they are not in bijection they can not be isomorphic.
Also the second property is false, indeed we would have

$$X \times (Y \times Z) \cong (X \times Y) \times (X \times Z)$$

that is false as we have proved above.

4. We have to prove that, given an equivalence of category $F : \mathcal{C} \to \mathcal{D}$, holds $F(A \prod B) \cong F(A) \prod F(B)$ and $F(A \coprod B) \cong F(A) \coprod F(B)$. Let's deal first with the product case. We recall that $F(A) \prod F(B)$ is the representative of the functor

$$\begin{array}{rccc} h^{F(A)} \times h^{F(B)} : & \mathcal{C} & \to & \mathbf{Set} \\ & Z & \mapsto & \mathrm{Hom}_{\mathcal{D}}(Z, F(A)) \times \mathrm{Hom}_{\mathcal{D}}(Z, F(B)) \end{array}$$

To prove that $F(A \prod B)$ is the representative of $h^{F(A)} \times h^{F(B)}$ we'll show that the following morphism of functor

$$\begin{array}{rccc} v_{h^{F(A)} \times h^{F(B)}, \xi} : & h^{F(A \prod B)} & \to & h^{F(A)} \times h^{F(B)} \\ & h^{F(A \prod B)}(T) & \to & h^{F(A)}(T) \times h^{F(B)}(T) \\ & (g : T \to F(A \prod B)) & \mapsto & (h^{F(A)} \times h^{F(B)})(g)(\xi) = g_*(\xi) = \xi \circ g \end{array}$$

is an isomorphism and that $\xi \in (h^{F(A)} \times h^{F(B)})(F(A \prod B))$ for any $T \in \mathrm{Ob}(\mathcal{D})$. Let's choose $\xi = (F(\pi_A), F(\pi_B))$ with π_A and π_B the projections given by the product $A \prod B$. Thus the morphism above became:

$$\begin{array}{rccc} v_{h^{F(A)} \times h^{F(B)}, \xi} : & h^{F(A \prod B)} & \to & h^{F(A)} \times h^{F(B)} \\ & h^{F(A \prod B)}(T) & \to & h^{F(A)}(T) \times h^{F(B)}(T) \\ & (g : T \to F(A \prod B)) & \mapsto & (F(\pi_A) \circ g, F(\pi_B) \circ g) \end{array}$$

From the theory we have that this is a morphism of functor. Let's prove also that each $v_{h^{F(A)} \times h^{F(B)}, \xi}(T)$ is bijective. Let's fix a pair (f, h) with $f : T \to F(A)$ and $h : T \to F(B)$. Since F is an equivalence of category we have that it is fully faithful and essentially surjective. Thus by essentially surjective-ness there exists $R \in \mathrm{Ob}(\mathcal{C})$ such that $F(R) = T$ and by fully faithful-ness we have unique morphisms $\overline{f} : R \to A$ and $\overline{h} : R \to B$ such that $F(\overline{f}) = f$ and $F(\overline{h}) = h$. Now, by the universal property of the product, we have a unique morphism $z_{f,h} : R \to A \prod B$ such that the following diagram

$$\begin{tikzcd} R \arrow[dr, "\exists!\, z_{f,h}"] \arrow[drr, bend left, "\overline{h}"] \arrow[ddr, bend right, "\overline{f}"'] & & \\ & A \prod B \arrow[r, "\pi_B"] \arrow[d, "\pi_A"] & B \\ & A & \end{tikzcd}$$

commutes. Thus the inverse map of $v_{h^{F(A)} \times h^{F(B)}, \xi}(T)$ is the map

$$\begin{array}{rccc} j(T) : & h^{F(A)}(T) \times h^{F(B)}(T) & \to & h^{F(A \prod B)}(T) \\ & (f, h) & \mapsto & F(z_{f,h}) \end{array}$$

this is well defined since $F(R) = T$ and since we have a unique map thanks to the universal properties. Now let's prove that it is the inverse:

$$\begin{aligned}(j(T) \circ v_{h^{F(A)} \times h^{F(B)}, \xi}(T))(g) &= j(T)(F(\pi_A) \circ g, F(\pi_B) \circ g) \\ &= F(z_{F(\pi_A)\circ g, F(\pi_B)\circ g})\end{aligned}$$

By construction, functoriality of F, the full faithfulness and the essential surjectivenes there exists $Q \in \mathcal{C}$ and $e : Q \to A \prod B$ such that $F(e) = g$ and we also have

$$F(\pi_A \circ e) = F(\pi_A) \circ g = F(\overline{F(\pi_A) \circ g}) = F(\pi_A) \circ F(z_{F(\pi_A)\circ g, F(\pi_B)\circ g}) = F(\pi_A \circ z_{F(\pi_A)\circ g, F(\pi_B)\circ g})$$

$$\implies \pi_A \circ e = \pi_A \circ z_{F(\pi_A)\circ g, F(\pi_B)\circ g}$$

and

$$F(\pi_B \circ e) = F(\pi_B) \circ g = F(\overline{F(\pi_B) \circ g}) = F(\pi_B) \circ F(z_{F(\pi_A)\circ g, F(\pi_B)\circ g}) = F(\pi_A \circ z_{F(\pi_A)\circ g, F(\pi_B)\circ g})$$

$$\implies \pi_B \circ e = \pi_B \circ z_{F(\pi_A)\circ g, F(\pi_B)\circ g}$$

Since $\pi_A \circ z_{F(\pi_A)\circ g, F(\pi_A)\circ g} = \overline{(\pi_A) \circ g}$ and $\pi_B \circ z_{F(\pi_A)\circ g, F(\pi_B)\circ g} = \overline{F(\pi_A) \circ g}$ we have just proved that the following diagram

$$\begin{tikzcd}
Q \arrow[dr, "e"] \arrow[drr, bend left, "\overline{F(\pi_B)\circ g}"] \arrow[ddr, bend right, "\overline{F(\pi_A)\circ g}"'] & & \\
& A\prod B \arrow[r, "\pi_B"] \arrow[d, "\pi_A"] & B \\
& A &
\end{tikzcd}$$

Now we use the universal property of the product: since also $z_{F(\pi_A)\circ g, F(\pi_A)\circ g}$ in place of e let the diagram commutes we have that $e = z_{F(\pi_A)\circ g, F(\pi_A)\circ g}$. Thus

$$g = F(e) = F(z_{F(\pi_A)\circ g, F(\pi_A)\circ g}) = (j(T) \circ v_{h^{F(A)} \times h^{F(B)}, \xi}(T))(g)$$

The other direction is easy since

$$\begin{aligned}(v_{h^{F(A)} \times h^{F(B)}, \xi}(T) \circ j(T))(f, h) &= v_{h^{F(A)} \times h^{F(B)}, \xi}(T)(F(z_{f,h})) \\ &= (F(\pi_A) \circ F(z_{f,h}), F(\pi_B) \circ F(z_{f,h})) \\ &= (F(\pi_A \circ z_{f,h}), F(\pi_B \circ z_{f,h})) \\ &= (F(\overline{f}), F(\overline{h})) = (f, h)\end{aligned}$$

Thus it is a bijective map, and we have proved the claim.
For the second statement we can observe that the product is the coproduct in the opposite category and that the functor F^{op} is still an equivalence of category. Thus we have

$$F(A) \coprod_{\mathcal{D}} F(B) = F(A) \prod_{\mathcal{D}^{op}} F(B) \cong F(A \prod_{\mathcal{C}^{op}} B) = F(A \coprod_{\mathcal{C}} B)$$

where the the isomorphism is true from the first part and since F^{op} is an equivalence between categories.

5. Let $H : \mathbf{Set} \to \mathbf{Set}^{op}$ an equivalence of category. Using the previous point of this exercise we have that for any $A, B : \mathrm{Ob}(\mathbf{Set})$ we have:

$$\begin{aligned}
H(A) \sqcup H(B) &= H(A) \coprod_{\mathbf{Set}} H(B) \\
&= H(A) \prod_{\mathbf{Set}^{op}} H(B) \\
&\cong H(A \prod_{\mathbf{Set}} B) \\
&= H(A \coprod_{\mathbf{Set}^{op}} B) \\
&\cong H(A) \coprod_{\mathbf{Set}^{op}} H(B) \\
&= H(A) \prod_{\mathbf{Set}} H(B) = H(A) \times H(B)
\end{aligned}$$

Moreover, since H is an equivalnce of category we have that it is essential surjective, thus there exist A and B such that $H(A) = \{1, 2, 3\} = H(B)$. Now we have a contradiction since $|H(A) \sqcup H(B)| = 6 \neq 9 = ||H(A) \times H(B)||$. Hence they are not isomorphic and then can not exist an quivalence of category such as H.

Exercise 20 (Exercise 4.5.8)**.** The goal of this exercise is to construct directly the categorical coproduct of two rings, without reference to the tensor product.

1. Let I and J be two sets. Construct the coproduct of $\mathbb{Z}[X_i]_{i\in I}$ and $\mathbb{Z}[Y_j]_{j\in J}$ in **Ring**. *Hint:* start by describing the functors represented by these two rings, then the functor that their coproduct should represent.

2. Let A and B be two rings. By choosing a generating set of A and a generating set of B, and using the first part, construct the coproduct of A and B in **Ring**.

Solution 20.

1. We have

$$h_{\mathbb{Z}[x]_{i\in I}} : \begin{array}{ccc} \mathbf{Ring} & \to & \mathbf{Set} \\ R & \mapsto & \mathrm{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x]_{i\in I}, R) \end{array}$$

Clearly we have $\mathrm{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x]_{i\in I}, R) \cong R^I$, and same holds for $h_{\mathbb{Z}[y]_{j\in J}}$. Thus we can study the functor:

$$h_{\mathbb{Z}[x]_{i\in I}} \times h_{\mathbb{Z}[y]_{j\in J}} : \begin{array}{ccc} \mathbf{Ring} & \to & \mathbf{Set} \\ R & \mapsto & h_{\mathbb{Z}[x]_{i\in I}}(R) \times h_{\mathbb{Z}[y]_{j\in J}}(R) \cong R^I \times R^J \cong R^{I\sqcup J} \end{array}$$

We clearly have that $h_{\mathbb{Z}[x]_{i\in I}} \times h_{\mathbb{Z}[y]_{j\in J}} \cong h_{\mathbb{Z}[z]_{k\in I\sqcup J}}$. Hence the above functor is represented by $\mathbb{Z}[z]_{k\in I\sqcup J}$; thus $(\mathbb{Z}[z]_{k\in I\sqcup J}, i, j)$, where i and j are the inclusion maps, is the coproduct in **Ring** between $\mathbb{Z}[X_i]_{i\in I}$ and $\mathbb{Z}[Y_j]_{j\in J}$.

2. Let's fix $A = \langle a_t\rangle_{t\in T,\, a_t\in A}$ $B = \langle b_s\rangle_{s\in S,\, b_s\in B}$ with A and B sets. Since we have surjective maps $f : \mathbb{Z}[x_t]_{t\in T} \to A$ and $g : \mathbb{Z}[y_s]_{s\in S} \to B$ obtained sending $x_i \mapsto a_i$ and $x_i \mapsto b_i$, we have $A \cong \mathbb{Z}[x_t]_{t\in T}/\mathfrak{a}$ and $B \cong \mathbb{Z}[x_s]_{s\in S}/\mathfrak{b}$ for some ideals $\mathfrak{a} \subset \mathbb{Z}[x_t]_{t\in T}$ and $\mathfrak{b} \subset \mathbb{Z}[x_s]_{s\in S}$. To find the coproduct we have to find the representative for the following functor:

$$h_A \times h_B : \begin{array}{ccc} \mathbf{Ring} & \to & \mathbf{Set} \\ R & \mapsto & h_A(R) \times h_B(R) \end{array}$$

that is isomorphic to

$$h_{\mathbb{Z}[x_t]_{t\in T}/\mathfrak{a}} \times h_{\mathbb{Z}[x_s]_{s\in S}/\mathfrak{b}} : \begin{array}{ccc} \mathbf{Ring} & \to & \mathbf{Set} \\ R & \mapsto & h_{\mathbb{Z}[x_t]_{t\in T}/\mathfrak{a}}(R) \times h_{\mathbb{Z}[x_s]_{s\in S}/\mathfrak{b}}(R) \end{array}$$

Now we have that

$$h_{\mathbb{Z}[x_t]_{t\in T}/\mathfrak{a}}(R) = \mathrm{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x_t]_{t\in T}/\mathfrak{a}, R) = \{(r_t)_{t\in T} \in R^T \mid \forall p \in \mathfrak{a},\, p((r_t)_{t\in T}) = 0\}$$

Since the same holds also for $h_{\mathbb{Z}[x_s]_{s\in S}/\mathfrak{b}}(R)$ and since we have

$$\begin{aligned} h_{\mathbb{Z}[x_k]_{k\in T\sqcup S}/(\mathfrak{a},\mathfrak{b})}(R) =& \mathrm{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x_k]_{k\in T\sqcup S}/(\mathfrak{a},\mathfrak{b}), R) \\ \cong& \left\{ (r_k)_{k\in S\sqcup T} \in R^{S\sqcup T} \;\middle|\; \begin{array}{l} \forall p \in \mathfrak{a},\, p((r_t)_{t\in T}) = 0 \\ \forall q \in \mathfrak{b},\, q((r_s)_{s\in S}) = 0 \end{array} \right\} \end{aligned}$$

we have that

$$\begin{aligned}
h_{\mathbb{Z}[x_t]_{t\in T}/\mathfrak{a}}(R) \times h_{\mathbb{Z}[x_s]_{s\in S}/\mathfrak{b}}(R) \cong & \{(r_t)_{t\in T} \in R^T \mid \forall p \in \mathfrak{a},\, p((r_t)_{t\in T}) = 0\} \\
& \times \\
& \{(r_s)_{s\in S} \in R^S \mid \forall p \in \mathfrak{b},\, p((r_s)_{s\in S}) = 0\} \\
\cong & \left\{ (r_k)_{k\in S\sqcup T} \in R^{S\sqcup T} \,\middle|\, \begin{matrix} \forall p \in \mathfrak{a},\, p((r_t)_{t\in T}) = 0 \\ \forall q \in \mathfrak{b},\, q((r_s)_{s\in S}) = 0 \end{matrix} \right\} \\
\cong & h_{\mathbb{Z}[x_k]_{k\in T\sqcup S}/(\mathfrak{a},\mathfrak{b})}(R)
\end{aligned}$$

Thus, defining

$$A \coprod B := \mathbb{Z}[x_k]_{k\in T\sqcup S}/(\mathfrak{a}, \mathfrak{b})$$

as object in **Ring** with the operations component by component, we have that $h_{A\coprod B} \cong h_A \times h_B$ as functors; thus we have that $(A \coprod B, i, j)$, where i and j are the obvious inclusion of A and B in $A \coprod B$, is the coproduct between A and B.

5 Limits and colimits

Exercise 21 (Exercise 5.1.8)**.** Let $\mathcal{C}$ be a category such that $\mathrm{Hom}(X, Y)$ has at most one element for every pair (X, Y) of objects of $\mathcal{C}$. Let $\mathcal{D} = ((X_i), E_{ij})$ be a diagram in $\mathcal{C}$, and let $\mathcal{D}'$ be the diagram consisting solely of the objects X_i, with no arrows. Show that $h^{\mathcal{D}} = h^{\mathcal{D}'}$ and $h_{\mathcal{D}} = h_{\mathcal{D}'}$.

Solution 21. We prove $h^{\mathcal{D}} = h^{\mathcal{D}'}$; the case $h_{\mathcal{D}} = h_{\mathcal{D}'}$ is dual and we give it at the end. Fix $Z \in \mathrm{Ob}(\mathcal{C})$. By definition, an element of $h^{\mathcal{D}}(Z) = \mathrm{Hom}(Z, \mathcal{D})$ is a *cone* from Z to the diagram $\mathcal{D}$, i.e. a family

$$(\varphi_i : Z \to X_i)_{i \in I}$$

of morphisms in $\mathcal{C}$ such that, for every $i, j \in I$ and every arrow $e \in E_{ij}$ (so $e : X_i \to X_j$ in the diagram), the following triangle commutes:

$$\begin{tikzcd} & X_i \arrow[dd, "e"] \\ Z \arrow[ur, "\varphi_i"] \arrow[dr, "\varphi_j"'] & \\ & X_j \end{tikzcd} \qquad \text{i.e.} \qquad e \circ \varphi_i = \varphi_j.$$

On the other hand, since $\mathcal{D}'$ has the same objects X_i and no arrows, an element of $h^{\mathcal{D}'}(Z)$ is simply a family

$$(\varphi_i : Z \to X_i)_{i \in I}$$

with *no* compatibility condition. Thus, as subsets of $\prod_{i \in I} \mathrm{Hom}(Z, X_i)$,

$$h^{\mathcal{D}}(Z) \subseteq h^{\mathcal{D}'}(Z).$$

We claim this inclusion is an equality, i.e. that every family $(\varphi_i)_{i \in I} \in h^{\mathcal{D}'}(Z)$ automatically satisfies the compatibility condition above, and hence already lies in $h^{\mathcal{D}}(Z)$. Indeed, fix $i, j \in I$ and $e \in E_{ij}$. Both

$$e \circ \varphi_i \quad \text{and} \quad \varphi_j$$

are elements of $\mathrm{Hom}(Z, X_j)$. By hypothesis, $|\mathrm{Hom}(Z, X_j)| \leq 1$; since φ_j itself belongs to $\mathrm{Hom}(Z, X_j)$, this set is nonempty, hence has *exactly* one element. Therefore

$$e \circ \varphi_i = \varphi_j,$$

as both sides are elements of a singleton. Since i, j, e were arbitrary, $(\varphi_i)_{i \in I}$ satisfies all the compatibility conditions of $\mathcal{D}$, i.e. $(\varphi_i)_{i \in I} \in h^{\mathcal{D}}(Z)$. This proves

$$h^{\mathcal{D}}(Z) = h^{\mathcal{D}'}(Z) \qquad \text{for every } Z \in \mathrm{Ob}(\mathcal{C}).$$

It remains to check naturality in Z, i.e. that this equality of sets defines an equality of functors $h^{\mathcal{D}} = h^{\mathcal{D}'} : \mathcal{C} \to \mathbf{Set}$. But for any morphism $g : Z' \to Z$ in $\mathcal{C}$, the induced maps

$$h^{\mathcal{D}}(g) : h^{\mathcal{D}}(Z) \to h^{\mathcal{D}}(Z'), \qquad h^{\mathcal{D}'}(g) : h^{\mathcal{D}'}(Z) \to h^{\mathcal{D}'}(Z')$$

are both given by precomposition, $(\varphi_i)_{i\in I} \mapsto (\varphi_i \circ g)_{i\in I}$, which is literally the same map once we identify $h^{\mathcal{D}}(Z) = h^{\mathcal{D}'}(Z)$ and $h^{\mathcal{D}}(Z') = h^{\mathcal{D}'}(Z')$ as above. Hence $h^{\mathcal{D}}(g) = h^{\mathcal{D}'}(g)$ under this identification, and the two functors coincide:

$$h^{\mathcal{D}} = h^{\mathcal{D}'}.$$

Exercise 22 (Exercise 5.1.9)**.** Let Ω be an ordered set, viewed as a category. Let $\mathcal{D} = (\omega_i)_i$ be a diagram in Ω (without arrows: the previous exercise shows that for any question concerning limits or colimits in Ω, one may restrict to diagrams without arrows). Under what condition, expressed in terms of properties of the ordered set Ω, does the diagram $\mathcal{D}$ admit a limit (resp. a colimit), and what is it in that case?

Solution 22. Let's deal the limit case, the colimit case is analogous. We have that $h^{\mathcal{D}} = \prod_{i \in I} h^{w_i}$. Thus we have to study for wich hypotesis of Ω the functor

$$\begin{array}{ccc} \Omega & \to & \mathbf{Set} \\ k & \mapsto & \prod_{i \in I} \mathrm{Hom}_\Omega(k, w_i) \end{array}$$

is representable. Hence, we have to find an element $L \in \Omega$ such that

$$\mathrm{Hom}_\Omega(k, L) \cong \prod_{i \in I} \mathrm{Hom}_\Omega(k, w_i)$$

in the natural way. From the definition of Ω as a category, we have that each homset is not bigger than a singleton; thus also $\prod_{i \in I} \mathrm{Hom}_\Omega(k, w_i)$ is not bigger then a singleton. Now let's suppose that for any subset A of Ω, A does admit a infimum. Choosing $L = \inf_{i \in I}\{w_i\}$ we clearly have the above isomorhpism (we have that the set on the left has cardinality 1 (resp. 0) if and only if the one on the right has cardinality 1 (resp. 0)). For any $j \leq k$ in Ω the diagram

$$\begin{tikzcd} \mathrm{Hom}_\Omega(k, L) \arrow[r] \arrow[d] & \mathrm{Hom}_\Omega(j, L) \arrow[d] \\ \prod_{i \in I} \mathrm{Hom}_\Omega(k, w_i) \arrow[r] & \prod_{i \in I} \mathrm{Hom}_\Omega(j, w_i) \end{tikzcd}$$

commutes (they are not bigger then a singleton!), thus we have that $h^{\mathcal{D}} \cong h^L$ and we have shown that the representative is L. Let's prove that the hypothesis on Ω is not restrictive. Let's consider the following ordered set

$$\begin{tikzcd} A & B \\ C \arrow[u] \arrow[ur] & D \arrow[u] \arrow[ul] \end{tikzcd}$$

with $\Omega = \{A, B, C, D\}$ and let's fix $\mathcal{D} = \{A, B\}$. To prove that the limit does not exists we can try all the for elements and prove that none of them satisfy the condition:

A: $\{*\} = \mathrm{Hom}_\Omega(A, A) \ncong \prod_{i \in I} \mathrm{Hom}_\Omega(A, w_i) = \mathrm{Hom}_\Omega(A, A) \prod \mathrm{Hom}_\Omega(A, B) = \emptyset$

B: $\{*\} = \mathrm{Hom}_\Omega(B, B) \ncong \prod_{i \in I} \mathrm{Hom}_\Omega(B, w_i) = \mathrm{Hom}_\Omega(B, A) \prod \mathrm{Hom}_\Omega(B, B) = \emptyset$

C: $\emptyset = \mathrm{Hom}_\Omega(D, C) \ncong \prod_{i \in I} \mathrm{Hom}_\Omega(D, w_i) = \mathrm{Hom}_\Omega(D, A) \prod \mathrm{Hom}_\Omega(D, B) = \{*\}$

D: $\emptyset = \mathrm{Hom}_\Omega(C, D) \ncong \prod_{i \in I} \mathrm{Hom}_\Omega(C, w_i) = \mathrm{Hom}_\Omega(C, A) \prod \mathrm{Hom}_\Omega(C, B) = \{*\}$

Thus there not exists the limit. Dealing with the dual case, once switched inf with sup, is easy.

Exercise 23 (Exercise 5.2.10)**.** Let (A_i) be an arbitrary family of rings. Construct their coproduct directly, without using the tensor product, by a method analogous to that of Exercise 20.

Solution 23.

Exercise 24 (Exercise 5.2.12)**.** Let **EnsPt** be the category of *pointed sets*. Its objects are pairs (X, x) formed by a set X and an element $x \in X$; a morphism from (X, x) to (Y, y) is a map $f\colon X \to Y$ such that $f(x) = y$. If (X, x) and (Y, y) are two pointed sets, show that their coproduct exists in **EnsPt** and describe it.

Solution 24. Let's consider two pointed sets (X, x) and (Y, y). Let's define $U := (X \sqcup Y)/\rho$, where the only elements in reletion are $x\rho y$. Let's prove that U with the obvious inclusions is the coproduct. In order to do this, let's consider a pointed set (Z, z) and morphisms $f : (X, x) \to (Z, z)$, $g : (Y, y) \to (Z, z)$. We have the following diagram

$$\begin{tikzcd}
(Z,z) & & \\
& (U,[x]) & (X,x) \arrow[l, "j"] \arrow[ull, bend right, "f"'] \\
& (Y,y) \arrow[u, "i"] \arrow[uul, bend left, "g"] &
\end{tikzcd}$$

Let's define the following map

$$\begin{array}{rccc}
h: & (U,[x]) & \to & (Z,z) \\
& [u] & \mapsto & \begin{cases} z, & \text{if } u = [x] \\ f(u), & \text{if } u \in X \\ g(u), & \text{if } u \in Y \end{cases}
\end{array}$$

This map is clearly well defined, since the relation ρ identify only x and y and we clearly have that the above diagram commutes. Now let's suppose that $k : (U, [x]) \to (Z, z)$ is another morphism such that the above diagram commute. It is clear that

$$k([x]) = z = h([x])$$

Now, if $u \in X$ we have

$$k([u]) = g(u) = h([u)]$$

and if $u \in Y$ we have

$$k([u]) = f(u) = h([u)]$$

Thus k is unique and $(U, [x])$ is the coproduct.

Exercise 25 (Exercise 5.3.7)**.** Let I be the set of strictly positive integers, equipped with the divisibility relation; this is a filtered ordered set. For every pair of strictly positive integers n and m such that $n \mid m$, denote by f_{nm} the unique group homomorphism from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{Z}/m\mathbb{Z}$ sending $\bar{1} \bmod n$ to $\overline{(m/n)} \bmod m$ (justify its existence). Let D be the colimit in **Ab** of the filtered commutative diagram consisting of the groups $\mathbb{Z}/n\mathbb{Z}$ for $n > 0$ and the morphisms f_{nm}.

Show that D is isomorphic to the subgroup of $\mathbb{C}^\times$ consisting of roots of unity. Show also, if possible directly (without using the first question), that D is isomorphic to $\mathbb{Q}/\mathbb{Z}$.

Solution 25. First of all let's see the existence for f_{nm}. It is clear that $\mathrm{ord}_{\mathbb{Z}/m\mathbb{Z}}(\overline{\frac{m}{n}}) = n$, thus we have that there exists a homomorphism that sends $\bar{1} \mapsto (\overline{\frac{m}{n}})$. Since the group $Z/n\mathbb{Z}$ is cyclic we have the existence and the uniqueness of the map f_{nm}.
Now let's recall that $\mu_n := \{e^{\frac{2k\pi \mathrm{i}}{n}} \mid k \in \{0, \dots n-1\}\}$ and let's call the groups of roots of unity $\mu_\infty = \bigcup_{n\in\mathbb{N}} \mu_n$. We know that the $D = \mathrm{colim}\mathcal{D}$ is

$$\left(\bigsqcup_{n\in\mathbb{N}} \mathbb{Z}/n\mathbb{Z}\right) / \sim$$

where $([a]_k \sim [b]_t)$ if and only if there exists $j \in \mathbb{N}$ with k and t dividing j such that $\left[a\frac{j}{k}\right]_j = \left[b\frac{j}{t}\right]_j$. We want to prove that the following function

$$\begin{array}{cccc} f: & D & \to & \mu_\infty \\ & \overline{[a]_n} & \mapsto & e^{\frac{2\pi a \mathrm{i}}{n}} \end{array}$$

is well defined. Thus lets consider an element $\overline{[a]_n} = \overline{[b]_m}$, thus we have $f(\overline{[a]_n}) = e^{\frac{2\pi a\mathrm{i}}{n}}$ and $f(\overline{[b]_m}) = e^{\frac{2\pi b\mathrm{i}}{m}}$. Since we have $\overline{[a]_n} = \overline{[b]_m}$ there exists a $j \in \mathbb{N}$ with n and m dividing j such that $\left[a\frac{j}{n}\right]_j = \left[b\frac{j}{m}\right]_j$; thus we have $0 = \left[a\frac{j}{n} - b\frac{j}{m}\right]_j = \left[\frac{ajm-bjn}{mn}\right]_j = \left[\frac{j}{mn}(am-bn)\right]_j$. Hence we have that j divides $\frac{j}{mn}(am-bn)$ and that nm divides $(am-bn)$. Thus there exists an integer $h \in \mathbb{Z}$ such that $am - bn = hnm$ and dividing by nm we get $\frac{a}{n} - \frac{b}{m} = h$ (thus $\frac{a}{n} = h + \frac{b}{m}$). Thus we have that

$$f(\overline{[a]_n}) = e^{\frac{2\pi a\mathrm{i}}{n}} = e^{2\pi\mathrm{i}(h+\frac{b}{m})} = e^{2\pi\mathrm{i}h + \frac{2\pi b\mathrm{i}}{m}} = e^{\frac{2\pi b\mathrm{i}}{m}} = f(\overline{[b]_m})$$

Thus the function is well defined. Now we have to prove that it is a group homomorphism. Given two element $\overline{[a]_n}$, $\overline{[b]_m}$ the operations are defined using the filtration of the index set.

Thus we have

$$\begin{aligned}
f(\overline{[a]_n}+\overline{[b]_m}) &= f\left(\overline{\left[\frac{anm}{n}+\frac{bnm}{m}\right]_{nm}}\right)\\
&= f\left(\overline{[am+bn]_{nm}}\right)\\
&= e^{\frac{2\pi\mathrm{i}(am+bn)}{nm}}\\
&= e^{\frac{2\pi\mathrm{i}am}{nm}+\frac{2\pi\mathrm{i}bn}{nm}}\\
&= e^{\frac{2\pi\mathrm{i}a}{n}+\frac{2\pi\mathrm{i}b}{m}}\\
&= e^{\frac{2\pi\mathrm{i}a}{n}}e^{\frac{2\pi\mathrm{i}b}{m}}\\
&= f(\overline{[a]_n})f(\overline{[b]_m})
\end{aligned}$$

(the group structure in μ_∞ is given by the multiplication in the complex field). Now we only have to prove that is bijective. Thus let's say that $f(\overline{[a]_n}) = 0_{\mu_\infty}$ for some $\overline{[a]_n} \in D$ (let's recall that $0_{\mu_\infty} = 1_{\mathbb{C}}$). Thus we have that

$$e^{\frac{2a\pi\mathrm{i}}{n}} = 1$$

then n divides a and we have $[a]_n = 0$. Thus it is injective. The surjectivity is easy since given root of unity $e^{\frac{2k\pi\mathrm{i}}{n}}$ we have $\overline{[k]_n} \in D$ such that $f(\overline{[k]_n}) = e^{\frac{2k\pi\mathrm{i}}{n}}$. Thus the groups are isomorphic.

Now let's build the ismorphism $D \cong \mathbb{Q}/\mathbb{Z}$. Let's prove that the map

$$\begin{array}{rccc}
f: & D & \to & \mathbb{Q}/\mathbb{Z}\\
 & \overline{[a]_n} & \mapsto & \left[\frac{a}{n}\right]
\end{array}$$

is well defined. Let's consider $\overline{[a]_n} = \overline{[b]_m}$, thus as we have done before, we have that exists an integer $h \in \mathbb{Z}$ such that $\frac{a}{n} = h + \frac{b}{m}$. Then we have that

$$f(\overline{[a]_n}) = \left[\frac{a}{n}\right] = \left[h+\frac{b}{m}\right] = \left[\frac{b}{m}\right] = f(\overline{[b]_m})$$

and the function is well defined. Moreover, it is also a group homomorphism since we have

$$\begin{aligned}
f(\overline{[a]_n}+\overline{[a]_n}) &= f\left(\overline{\left[\frac{anm}{n}+\frac{bnm}{m}\right]_{nm}}\right)\\
&= f\left(\overline{[am+bn]_{nm}}\right)\\
&= \left[\frac{am+bn}{nm}\right]\\
&= \left[\frac{am}{nm}\right]+\left[\frac{bn}{nm}\right]\\
&= \left[\frac{a}{n}\right]+\left[\frac{b}{m}\right]\\
&= f(\overline{[a]_n})+f(\overline{[b]_m})
\end{aligned}$$

Now let's prove that is bijective. Thus let's say that $f(\overline{[a]_n}) = 0_{\mathbb{Q}/\mathbb{Z}}$ for some $\overline{[a]_n} \in D$; hence we have that $[0] = f(\overline{[a]_n}) = \left[\frac{a}{n}\right]$ and we have that $n|a$, thus we have also that $[a]_n = 0$. Then f has trivial kernel and is injective. Again, the surjectivity is easy since given a $\left[\frac{a}{b}\right] \in \mathbb{Q}/\mathbb{Z}$ we have that $f(\overline{[a]_b}) = \left[\frac{a}{b}\right]$. Thus we have the following isomorphisms

$$\mu_\infty \cong D \cong \mathbb{Q}/\mathbb{Z}$$

6 Adjocntion

Exercise 26 (Exercise 6.3.3)**.** Let $\mathcal{D}$ be the filtered commutative diagram in **Compl** indexed by $\{n \in \mathbb{N},\, n \geq 2\}$, whose family of objects is $\left([\frac{1}{n}, 1]\right)_{n\geq 2}$ and whose arrows are the inclusion maps.

1. Show that the colimit of $\mathcal{D}$ in **Met** exists and equals $]0, 1]$ (equipped with the inclusion maps $[\frac{1}{n}, 1] \subset]0, 1]$).

2. Show that the colimit of $\mathcal{D}$ in **Compl** exists and equals $[0, 1]$ (equipped with the inclusion maps $[\frac{1}{n}, 1] \subset [0, 1]$).

3. Does the inclusion functor from **Compl** to **Met** admit a right adjoint?

Solution 26.

1. Let $f_k : X_k \to Z$ be maps such that

$$\begin{tikzcd} X_k \arrow[rr, "k"] \arrow[dr, "f_k"'] & & X_j \arrow[dl, "f_j"] \\ & Z & \end{tikzcd} \tag{1}$$

the above diagram commutes. Let's build a morphism $a : (0, 1] \to Z$. Let $x \in (0, 1]$, thus there exists $n_x \in \mathbb{N}$ such that $x \in \left[\frac{1}{n_x}, 1\right]$. Hence we can define

$$\begin{array}{rccc} a: & (0,1] & \to & Z \\ & x & \mapsto & f_{n_x}(x) \end{array}$$

Since the diagram (2) commutes, we have that the function a is well defined. By construction we also have that $f_j = a \circ i_j$, thus we have the following commutative diagram

$$\begin{tikzcd} X_k \arrow[rr, "i_{kj}"] \arrow[dr, "i_k"] \arrow[ddr, "f_k"'] & & X_j \arrow[dl, "i_j"'] \arrow[ddl, "f_j"] \\ & (0,1] \arrow[d, "a"] & \\ & Z & \end{tikzcd}$$

Now we want to prove that a is the unique map. Let $g : (0, 1] \to Z$ another function such that the above diagram with g instead of a commutes. Let $x \in (0, 1]$, as above, there exists $n_x \in \mathbb{N}$ such that $x \in \left[\frac{1}{n_x}, 1\right]$. Thus we have that $g(x) = f_{n_x}(x) = a \circ i_{n_x}(x) = a(x)$. Then $(0, 1]$ is the colimit.

2. The proof is analogous to the previous one, it suffices to replace $(0, 1]$ with $[0, 1]$ since the last one is complete ($(0, 1]$ it's not!).

3. The exercise asks if there exists a functor $G : \mathbf{Met} \to \mathbf{Compl}$ such that for any $X \in \mathrm{Ob}(\mathbf{Compl})$ and $Y \in \mathrm{Ob}(\mathbf{Met})$ we have bijections between

$$\mathrm{Hom}_{\mathbf{Met}}(i(X), Y) \cong \mathrm{Hom}_{\mathbf{Compl}}(X, G(Y))$$

Let's suppose that there exists a such functor G, then we have that i is a left adjoint and then it should preserve the colimits. In the previous point we have shown that the same diagram gives two differents colimit, thus it does not preserve the diagrams and then cannot exists a right adjoint.

Exercise 27 (Exercise 6.3.6.)**.** Let O_5 be the forgetful functor from **EnsPt** to **Ens** (the category **EnsPt** was defined in Exercise 5.2.12, cfr.(24)). Show that O_5 has a left adjoint, which will be constructed explicitly. Show that it does not have a right adjoint.

Solution 27. In order to find the left adjoint functor, we have to search for a functor $G : \mathbf{Ens} \to \mathbf{EnsPt}$ such that for any $X \in \mathrm{Ob}(\mathbf{Ens})$ and $(Y, y) \in \mathrm{Ob}(\mathbf{EnsPt})$

$$\mathrm{Hom}_{\mathbf{EnsPt}}(G(X), (Y, y)) \cong \mathrm{Hom}_{\mathbf{Ens}}(X, O_5(Y, y))$$

that is equivalent to

$$\mathrm{Hom}_{\mathbf{EnsPt}}(G(X), (Y, y)) \cong \mathrm{Hom}_{\mathbf{Ens}}(X, Y)$$

We can define

$$\begin{array}{rccc} G: & \mathbf{Set} & \to & \mathbf{SetPt} \\ & X & \mapsto & (X \sqcup \{\star\}, \star) \end{array}$$

and the isomorphism is

$$\begin{array}{rccc} k: & \mathrm{Hom}_{\mathbf{EnsPt}}(G(X), (Y, y)) & \to & \mathrm{Hom}_{\mathbf{Ens}}(X, Y) \\ & f & \mapsto & f_{|X} \end{array}$$

Let's suppose that O_5 admits a right adjoint functor, it means that O_5 should be a left adjoint functor. Thus, we should have that O_5 commutes with the colimits (and then also with the coproducts). Instead, using the exercise 24, and defining $\mathbb{A} := \{1, 2\}$ we have

$$O_5\left((\mathbb{A}, 1) \coprod (\mathbb{A}, 1)\right) = O_5((\mathbb{A} \sqcup \mathbb{A})/\rho, [1]) = (\mathbb{A} \sqcup \mathbb{A})/\rho \ncong \mathbb{A} \sqcup \mathbb{A} = O_5(\mathbb{A}, 1) \coprod O_5(\mathbb{A}, 1)$$

Where $(\mathbb{A} \sqcup \mathbb{A})/\rho \ncong \mathbb{A} \sqcup \mathbb{A}$ since they have different cardinality. Since the forgetful functor does not commutes with the coprodcuts (and thus also with the colimits) we have that cannot admit a right adjoint functor.

7 Partial exam

Exercise 28 (Exercise 1)**.** Let $\mathcal{C}$ be the category of finite groups, viewed as a full subcategory of **Gp**. Let F be the forgetful functor from $\mathcal{C}$ to **Set**. Show that F is not representable.

Solution 28. let's suppose that $F : \mathcal{C} \to \mathbf{Set}$ is representable, then there exists a finite group X such that we have a bijection for any group G

$$G = F(G) \cong \mathrm{Hom}_{\mathbf{Gp}}(X, G) = \mathrm{Hom}_{\mathcal{C}}(X, G)$$

Choosing $G = \mathbb{Z}/p\mathbb{Z}$, we have that

$$p = |\mathbb{Z}/p\mathbb{Z}| = |\mathrm{Hom}_{\mathcal{C}}(X, \mathbb{Z}/p\mathbb{Z})|$$

If $f \in \mathrm{Hom}_{\mathcal{C}}(X, \mathbb{Z}/p\mathbb{Z})$ then there exists an element $x \in X$ such that $p|\mathrm{ord}_X(x)$. Since for any p we have

$$|\mathrm{Hom}_{\mathcal{C}}(X, \mathbb{Z}/p\mathbb{Z})| = p$$

then, for any p we have at least one element in X of order p. This means that X has non-finite cardinality. Then. F is not representable.

Exercise 29 (Exercise 2)**.** Soit $F\colon \mathbf{Gp} \to \mathbf{Ens}$ le foncteur d'oubli. Décrire entièrement $\mathrm{End}\, F$ et $\mathrm{Aut}\, F$.

Solution 29. First of all, we can observe that F is representable by the group $\mathbb{Z}$. Thus, we will write $F = h_{\mathbb{Z}}$.

$\mathrm{End}(F)$: By definition, a natural transformation $\eta : h_{\mathbb{Z}} \implies h_{\mathbb{Z}}$ is an endomorphism for $h_{\mathbb{Z}}$. Using the Yoneda lemma, we have that η is linked to an unique element in $h_{\mathbb{Z}}(\mathbb{Z})$. More precisely, the Yoneda lemma says that,

$$\mathrm{End}(F) := \{u_{h_{\mathbb{Z}},f} \mid f \in h_{\mathbb{Z}}(\mathbb{Z}) = \mathrm{Hom}_{\mathbf{Gp}}(\mathbb{Z}, \mathbb{Z})\}$$

where

$$\begin{array}{rccc} u_{h_{\mathbb{Z}},f} : & h_{\mathbb{Z}} & \implies & h_{\mathbb{Z}} \\ u_{h_{\mathbb{Z}},f}(G) : & h_{\mathbb{Z}}(G) & \to & h_{\mathbb{Z}}(G) \\ & (a : \mathbb{Z} \to G) & \mapsto & h_{\mathbb{Z}}(f)(a) := a \circ f \end{array}$$

Moreover, since since a morphism from $\mathbb{Z} \to \mathbb{Z}$ it's defined from the immage of 1, we can write each f as

$$\begin{array}{rccc} f_i : & \mathbb{Z} & \to & \mathbb{Z} \\ & 1 & \mapsto & i \end{array}$$

Then we have

$$\mathrm{End}(F) := \{u_{h_{\mathbb{Z}},f_i} \mid i \in \mathbb{Z}\}$$

$\mathrm{Aut}(F)$: Now, we want to find the automorphism for F, and we can also search them in $\mathrm{End}(F)$. We know that a natural transformation $u_{h_{\mathbb{Z}},f_i}$ is an isomorphism, if and only if f is an isomorphism. This is true if and only if $i \in \{1, -1\}$. Then we have

$$\mathrm{Aut}(F) = \{u_{h_{\mathbb{Z}},f_1}, u_{h_{\mathbb{Z}},f_{-1}}\}$$

Exercise 30 (Exercise 3)**.** Let G be a group and let C be the category of left G-sets, defined as follows: an object of C is a set X equipped with a left action $(g, x) \mapsto gx$ of G on X; a morphism from X to Y in C is a map $f\colon X \to Y$ that is *equivariant*, i.e. $f(gx) = gf(x)$ for all $(g, x) \in G \times X$. We denote by $F\colon \mathsf{C} \to \mathbf{Set}$ the forgetful functor.

1. Show that a morphism in C is an isomorphism if and only if it is bijective as a map of sets.

2. Let $g \in G$. For every object X of C, let $\iota_g(X)$ denote the bijection $x \mapsto gx$ from X to X. Verify carefully that the collection of maps $\iota_g(X)$ defines a natural transformation ι_g from F to itself. What is $\iota_g \circ \iota_h$ (for $g, h \in G$)? Deduce that each ι_g is an automorphism of the functor F, and give its inverse.

3. Show that F is representable. If (R, r) denotes a representing object of F, describe the automorphism of R that corresponds to ι_g via the Yoneda lemma.

4. Show that $g \mapsto \iota_g$ defines a group isomorphism between G and $\operatorname{Aut} F$, and that $\operatorname{End} F = \operatorname{Aut} F$.

Solution 30.

1. Let $f \in \operatorname{Hom}_{\mathcal{C}}(X, Y)$ be an isomorphism, thus exists $g \in \operatorname{Hom}_{\mathcal{C}}(Y, X)$ with $g \circ f = \operatorname{Id}_X$ and $f \circ g = \operatorname{Id}_Y$. Since the composition law is the composition of functions, we have that f is a bijection of sets. Now let $f \in \operatorname{Hom}_{\mathcal{C}}(X, Y)$ a morphism such that f is bijective as map between maps. We have to prove that $h := f^{-1} : Y \to X$ is equivariant. Thus let $(g, y) \in G \times Y$, then we have
$$h(gy) = h(gf(f^{-1}(y))) = h(f(gf^{-1}(y))) = gf^{-1}(y) = gh(y)$$

2. Let's show that ι_g is a natural trnasformation. Let X, Y to G sets and $f \in \operatorname{Hom}_{\mathcal{C}}(X, Y)$. Let's consider the diagram
$$\begin{array}{ccc} X = F(X) & \xrightarrow{\iota_g(X)} & F(X) = X \\ \Big\downarrow{\scriptstyle F(f)} & & \Big\downarrow{\scriptstyle F(f)} \\ Y = F(Y) & \xrightarrow{\iota_g(Y)} & F(Y) = Y \end{array}$$
and $x \in X$ be an element, then we have
$$(F(f) \circ \iota_g(X))(x) = F(f)(gx) = f(gx) = gf(x) = \iota_g(Y)(f(x)) = (\iota_g(Y) \circ F(f))(x)$$
Thus the diagram commutes and it is a natural transformation. Moreover, let's fix a group X, then
$$(\iota_g \circ \iota_h)(X) = \iota_g(X) \circ \iota_h(X) : \begin{array}{ccc} X & \to & X \\ x & \mapsto & hgx \end{array}$$
It is clear that ι_g is an isomorphism, since it's inverse is $\iota_{g^{-1}}$.

3. In order to prove that (G, e_G) represents F, let's show the isomorphism
$$X = F(X) \cong h_G(X) = \mathrm{Hom}_{\mathcal{C}}(G, X)$$
for any G-set X. We can build the bijective map in the following way:
$$\begin{array}{rccc} \phi(X): & X & \to & \mathrm{Hom}_{\mathcal{C}}(G, X) \\ & x & \mapsto & \left(\begin{array}{rccc} f_x: & G & \to & X \\ & g & \mapsto & gx \end{array}\right) \end{array}$$
with inverse
$$\begin{array}{rccc} \phi^{-1}(X): & \mathrm{Hom}_{\mathcal{C}}(G, X) & \to & X \\ & f & \mapsto & f(e_G) \end{array}$$
Clearly they are bijective and we also have that

- $f_x(e_G) = e_G x = x$ from the property of the left action;
- $hf_x(g) = hgx = f_x(hg)$, thus it really is equivariant.

Moreover, using hte following corollary of the Yoneda lemma we have that
$$\begin{array}{rccc} \psi: & \mathrm{Hom}_{\mathcal{C}}(G, G) & \to & \mathrm{Nat}(F, F) \\ & f & \mapsto & u_{F,e_G} \circ f^* \circ u_{F,e_G}^{-1} = \phi^{-1} \circ f^* \circ \phi \end{array}$$
is a bijection. Thus, we want to find f such that
$$\phi^{-1} \circ f^* \circ \phi = \iota_g$$
Choosing f such that $f(h) := j_g(h) := gh$, we have that
$$\begin{array}{rccc} \phi^{-1}(X) \circ f^*(X) \circ \phi(X): & X & \to & X \\ & h & \mapsto & \phi^{-1}(X)(j_g^*(X)(f_h)) = \phi^{-1}(X)(f_h \circ j_g) = (f_h \circ j_g)(e_G) = f_h(g) = g \end{array}$$
Thus we have that
$$\phi^{-1}(X) \circ f^*(X) \circ \phi(X) = \iota_g$$
Then the natural transformation ι_g corresponds to the automorphism j_g.

4. First of all, we can observe that, since F is representable we have that $|\mathrm{Hom}(G, G)| = |\mathrm{Nat}(F, F)|$, thus $\mathrm{Nat}(F, F)$ is a set; then $\mathrm{Aut}(f)$ is a group with the composition. Let's consider the following homomorphism
$$\begin{array}{rccc} \eta: & G & \to & \mathrm{Aut}(F) \\ & g & \mapsto & \iota_g \end{array}$$
If $\iota_g = \mathrm{Id}_F$, then for any $x \in G$ we have $x = \mathrm{Id}_F(G)(x) = \iota_g(G)(x) = gx$, thus $g = e_G$. Then η is injective. Moreover, using the previous points, we have that
$$\mathrm{End}(F) = \mathrm{Nat}(F, F) \overset{\psi(G)}{\cong} \mathrm{Hom}_{\mathcal{C}}(G, G) \overset{\phi(G)}{\cong} G$$
In particular, an element g is sent into
$$g \mapsto \quad \phi^{-1} \circ f_g \circ \phi: \quad F \quad \to \quad F$$
and we have already proved in the previous point that such a natural transformation is of the form ι_g. Thus we have that, η is an isomorphism and that $\mathrm{End}(F) = \mathrm{Aut}(F)$.

Exercise 31 (Exercise 4)**.** Let C be a category and let $f : Y \to X$ be an arrow of C.

1. Recall that f is called a *monomorphism* (resp. an *epimorphism*) if for every object Z of C the map $\varphi \mapsto f \circ \varphi$ (resp. $\psi \mapsto \psi \circ f$) from $\mathrm{Hom}(Z, Y)$ to $\mathrm{Hom}(Z, X)$ (resp. from $\mathrm{Hom}(X, Z)$ to $\mathrm{Hom}(Y, Z)$) is injective. Reformulate these properties in terms of the functors h^X, h^Y, h_X and h_Y.

2. Assume that C admits fiber products. Let $\delta : Y \to Y \times_X Y$ be the *diagonal* of f, that is, the morphism corresponding to the pair $(\mathrm{Id}_Y, \mathrm{Id}_Y)$. By reasoning about the functors h^X and h^Y, show that f is a monomorphism if and only if δ is an isomorphism.

3. When C admits amalgamated sums, state (without proof) the dual assertion of the previous one.

4. We return to the assumption that C admits fiber products. Let $g : Z \to Y$ be a morphism; in what follows the notation $Z \times_X Z$ refers to the composite morphism $f \circ g : Z \to X$. Denote by u the morphism from $Z \times_Y Z$ to $Z \times_X Z$ induced by the two projections from $Z \times_Y Z$ to Z, and by v the morphism from $Z \times_Y Z$ to $Y \times_X Y$ defined by the pair (g, g). Justify that u and v are well defined, and recall the definition of the natural morphism w from $Z \times_Y Z$ to Y.

5. Show that the diagram

$$\begin{array}{ccc} Z \times_Y Z & \xrightarrow{u} & Z \times_X Z \\ {\scriptstyle w}\downarrow & & \downarrow{\scriptstyle v} \\ Y & \xrightarrow[\delta]{} & Y \times_X Y \end{array}$$

identifies $Z \times_Y Z$ as the fiber product of Y and $Z \times_X Z$ over $Y \times_X Y$ (with respect to the morphisms δ and v). Moreover, the canonic morphism w is the one given by the composition $w := g \circ \pi_1 = g \circ \pi_2$.

Solution 31.

1. Let $f : Y \to X$ be a monomorphism. Thus, the map

$$H(f) : \begin{array}{ccc} \mathrm{Hom}(Z, Y) & \to & \mathrm{Hom}(Z, X) \\ \phi & \mapsto & f \circ \phi \end{array}$$

is injective for any Z. We have that $\mathrm{Hom}(Z, Y) = h_Y(Z)$ and $\mathrm{Hom}(Z, X) = h_X(Z)$, thus the natural transformation

$$f_*(Z) : \begin{array}{ccc} h_Y(Z) & \to & h_X(Z) \\ \phi & \mapsto & f \circ \phi \end{array}$$

is injective for any Z. Similarly, f is an epimorphism if and only if the map

$$f^*(Z) : \begin{array}{ccc} h^X(Z) & \to & h^Y(Z) \\ \phi & \mapsto & \phi \circ f \end{array}$$

is injective for any Z.

2. We have the following commutative diagram

$$\begin{tikzcd}
Y \arrow[dr, "\delta"] \arrow[drr, bend left, "\mathrm{Id}_Y"] \arrow[ddr, bend right, "\mathrm{Id}_Y"'] & & \\
& Y \times_X Y \arrow[r, "\pi_1"] \arrow[d, "\pi_2"] & Y \arrow[d, "f"] \\
& Y \arrow[r, "f"] & X
\end{tikzcd}$$

Then, applying the functor $\mathrm{Hom}(Z, \cdot)$, from the Yoneda lemma we get that the above diagram commutes if and only if the following diagram commutes

$$\begin{tikzcd}
\mathrm{Hom}(Z,Y) \arrow[dr, "\delta_*"] \arrow[drr, bend left, "\mathrm{Id}_{Y,*}"] \arrow[ddr, bend right, "\mathrm{Id}_{Y,*}"'] & & \\
& \mathrm{Hom}(Z, Y \times_X Y) \arrow[r, "\pi_{1,*}"] \arrow[d, "\pi_{2,*}"] & \mathrm{Hom}(Z,Y) \arrow[d, "f_*"] \\
& \mathrm{Hom}(Z,Y) \arrow[r, "f_*"] & \mathrm{Hom}(Z,X)
\end{tikzcd}$$

From the naturality of the fiber product, instead of considering the above diagram, we can consider the following one

$$\begin{tikzcd}
\mathrm{Hom}(Z,Y) \arrow[dr, "\overline{\delta_*}"] \arrow[drr, bend left, "\mathrm{Id}_{Y,*}"] \arrow[ddr, bend right, "\mathrm{Id}_{Y,*}"'] & & \\
& \mathrm{Hom}(Z,Y) \times_{\mathrm{Hom}(Z,X)} \mathrm{Hom}(Z,Y) \arrow[r, "\pi_{1,*}"] \arrow[d, "\pi_{2,*}"] & \mathrm{Hom}(Z,Y) \arrow[d, "f_*"] \\
& \mathrm{Hom}(Z,Y) \arrow[r, "f_*"] & \mathrm{Hom}(Z,X)
\end{tikzcd}$$

Where $\pi_{i,*}$ are the classical projections, and the diagonal map is

$$\begin{array}{rccc}
\overline{\delta_*}: & \mathrm{Hom}(Z,Y) & \to & \mathrm{Hom}(Z,Y) \times_{\mathrm{Hom}(Z,X)} \mathrm{Hom}(Z,Y) \\
& \lambda & \mapsto & (\lambda, \lambda)
\end{array}$$

We clearly have that $\overline{\delta_*}$ is injective. Now, f is a monomorphism if and only if f_* is injective. Let $(a,b) \in \mathrm{Hom}(Z,Y) \times_{\mathrm{Hom}(Z,X)} \mathrm{Hom}(Z,Y)$, thus we have

$$f_*(b) = f_*(\pi_{2,*}(a,b)) = f_*(\pi_{1,*}(a,b)) = f_*(a)$$

Moreover, f_* is injective if and only if $b = a$. This means that the only element in $\mathrm{Hom}(Z,Y) \times_{\mathrm{Hom}(Z,X)} \mathrm{Hom}(Z,Y)$ are the ones of the form (λ, λ). Thus we have that

f_* is injective if and only if $\overline{\delta_*}$ is surjective. Since the injectivity for $\overline{\delta_*}$ is free we have that f is monomorphism if and only if $\overline{\delta_*}$ is an isomorphism. From the naturality of fiber product we have that $\overline{\delta_*}$ is an isomorphism if and only if δ_* is an isomporhism. Since we also have that δ_* is an isomorphsm if and only if δ is an isomorphism, we have done.

3. The dual statement is: "The map f is an epimorphism if and only if the map $\pi : Y \coprod_X Y \to Y$ induced by the maps $(\mathrm{Id}_Y, \mathrm{Id}_Y)$ is an isomorphism."

4. We have the following diagrams

$$\begin{tikzcd}
Z \times_Y Z \arrow[r, "\pi_2"] \arrow[d, "\pi_1"] & Z \arrow[d, "g"] \\
Z \arrow[r, "g"] & Y
\end{tikzcd}$$

$$\begin{tikzcd}
Z \times_Y Z \arrow[dr, "u"] \arrow[drr, bend left, "\pi_2"] \arrow[ddr, bend right, "\pi_1"'] & & \\
& Z \times_X Z \arrow[r, "\pi_4"] \arrow[d, "\pi_3"] & Z \arrow[d, "f\circ g"] \\
& Z \arrow[r, "f\circ g"] & X
\end{tikzcd}$$

$$\begin{tikzcd}
Z \times_Y Z \arrow[dr, "v"] \arrow[drr, bend left, "g\circ\pi_2"] \arrow[ddr, bend right, "g\circ\pi_1"'] & & \\
& Y \times_X Y \arrow[r, "\pi_6"] \arrow[d, "\pi_5"] & Y \arrow[d, "f"] \\
& Y \arrow[r, "f"] & X
\end{tikzcd}$$

From the universal property of the fiber product, we have that there exist unique maps u and v such that the above diagram commute. Thus, the two maps are well defined. Moreover, the canonic morphism w is the one given by the composition $w := g \circ \pi_1 = g \circ \pi_2$.

5. To get the claim, let's consider an object H with a couple of morphisms (a, b) such that the diagram with solid arrows

$$\begin{tikzcd}
H \arrow[dr, dotted, "k"] \arrow[drr, bend left, "b"] \arrow[ddr, bend right, "a"'] & & \\
& Z \times_Y Z \arrow[r, "u"] \arrow[d, "w"'] & Z \times_X Z \arrow[d, "v"] \\
& Y \arrow[r, "\delta"'] & Y \times_X Y
\end{tikzcd} \tag{2}$$

commute. Let's prove that there exists a unique morphism k such that the above diagram (with solid and dotted arrows now) commute. Applying again the Yoneda lemma, we gat that the above diagram is equivalent to the following diagram

$$\begin{tikzcd}
\mathrm{Hom}(T,H) \arrow[dr, dotted, "k_*"] \arrow[drr, bend left, "b_*"] \arrow[ddr, bend right, "a_*"'] & & \\
& \mathrm{Hom}(T, Z\times_Y Z) \arrow[r, "u_*"] \arrow[d, "w_*"'] & \mathrm{Hom}(T, Z\times_X Z) \arrow[d, "v_*"] \\
& \mathrm{Hom}(T,Y) \arrow[r, "\delta_*"'] & \mathrm{Hom}(T, Y\times_X Y)
\end{tikzcd}$$

That is also equivalent to

$$\begin{tikzcd}
\mathrm{Hom}(T,H) \arrow[dr, dotted, "\overline{k_*}"] \arrow[drr, bend left, "\overline{b_*}"] \arrow[ddr, bend right, "\overline{a_*}"'] & & \\
& \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,Y)}\mathrm{Hom}(T,Z) \arrow[r, "\overline{u_*}"] \arrow[d, "\overline{w_*}"'] & \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,X)}\mathrm{Hom}(T,Z) \arrow[d, "\overline{v_*}"] \\
& \mathrm{Hom}(T,Y) \arrow[r, "\overline{\delta_*}"'] & \mathrm{Hom}(T,Y)\times_{\mathrm{Hom}(T,X)}\mathrm{Hom}(T,Y)
\end{tikzcd}$$

Where the maps are

$$\begin{array}{rccc}
\overline{\delta_*}: & \mathrm{Hom}(T,Y) & \to & \mathrm{Hom}(T,Y)\times_{\mathrm{Hom}(T,X)}\mathrm{Hom}(T,Y) \\
& a & \mapsto & (a,a) \\
\\
\overline{v_*}: & \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,Z)}\mathrm{Hom}(T,Y) & \to & \mathrm{Hom}(T,Y)\times_{\mathrm{Hom}(T,X)}\mathrm{Hom}(T,Y) \\
& (a,b) & \mapsto & (g\circ a, g\circ b) \\
\\
\overline{u_*}: & \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,Y)}\mathrm{Hom}(T,Z) & \to & \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,X)}\mathrm{Hom}(T,Z) \\
& (a,b) & \mapsto & (a,b) \\
\\
\overline{w_*}: & \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,Y)}\mathrm{Hom}(T,Z) & \to & \mathrm{Hom}(T,Y) \\
& (a,b) & \mapsto & g\circ a = g\circ b
\end{array}$$

Thus we can define

$$\begin{array}{rccc}
\overline{k_*}: & \mathrm{Hom}(T,H) & \to & \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,Y)}\mathrm{Hom}(T,Z) \\
& a & \mapsto & \overline{b_*}(a)
\end{array}$$

To prove that is well defined, we have to show that for any $\lambda \in \mathrm{Hom}(T,H)$

$$\overline{b_*}(\lambda) \in \mathrm{Hom}(T,Z)\times_{\mathrm{Hom}(T,Y)}\mathrm{Hom}(T,Z)$$

which means that we have to show that

$$(g \circ p_1)(\overline{b_*}(\lambda)) = (g \circ p_2)(\overline{b_*}(\lambda))$$

(where p_i are the projections for $\mathrm{Hom}(T, Z) \times_{\mathrm{Hom}(T,Y)} \mathrm{Hom}(T, Z)$). Using the commutativity of the solid arrow diagram (2), we get that

$$[(g \circ p_1)((\overline{b_*}(\lambda))), (g \circ p_2)((\overline{b_*}(\lambda)))] = \overline{v_*}(\overline{b_*}(\lambda)) = \overline{\delta_*}(\overline{a_*}(\lambda)) = (\overline{a_*}(\lambda), \overline{a_*}(\lambda)) \qquad (3)$$

Thus, we have that the first couple has the same entrances, which means that

$$(g \circ p_1)((\overline{b_*}(\lambda))) = (g \circ p_2)((\overline{b_*}(\lambda)))$$

Thus is well defined. Now we have to prove that the diagram with $\overline{k_*}$ commutes.

- Since $\overline{u_*}$ sends $(a, a) \mapsto (a, a)$, from the definition $\overline{k_*} = \overline{b_*}$ we have that $\overline{k_*} \circ \overline{u_*} = \overline{b_*}$.
- Now we have to prove that $\overline{u_*} \circ \overline{k_*} = \overline{a_*}$. We have

  $$(\overline{w_*} \circ \overline{k_*})(\lambda) = (g \circ p_1 \circ \overline{k_*})(\lambda) = (g \circ p_1)(\overline{k_*}(\lambda)) = (g \circ p_1)(\overline{b_*}(\lambda))$$

  using the relation (3) we get

  $$(g \circ p_1)(\overline{b_*}(\lambda)) = \overline{a_*}(\lambda)$$

  Now we only have to prove that $\overline{k_*}$ is unique. This is easy since if A is another map as $\overline{k_*}$, we should have

  $$A(\lambda) = (\overline{u_*} \circ A)(\lambda) = \overline{b_*}(\lambda) = (\overline{u_*} \circ \overline{k_*})(\lambda) = \overline{k_*}(\lambda)$$

Exercise 32 (Exercise 5)**.** We will refer in what follows to the notions of monomorphism and epimorphism recalled in question (1) of the previous exercise. We denote by $\mathsf{Top}^{\mathrm{sep}}$ the full subcategory of Top consisting of *separated* (Hausdorff) topological spaces (recall that X is separated if and only if for every pair (x, y) of points of X with $x \neq y$ there exist two open sets U and V of X with $x \in U$, $y \in V$ and $U \cap V = \varnothing$).

1. Let f be the inclusion of $\mathbb{Q}$ into $\mathbb{R}$, viewed as a continuous map (where $\mathbb{Q}$ is equipped with the topology induced by that of $\mathbb{R}$). Show that f is an epimorphism in $\mathsf{Top}^{\mathrm{sep}}$, but that it is not an epimorphism in Top.

2. Let g be the map $t \mapsto \exp(2i\pi t)$ from $[0, 1[$ to the set U of complex numbers of modulus 1. Show that g is both a monomorphism and an epimorphism in $\mathsf{Top}^{\mathrm{sep}}$, but is not an isomorphism in $\mathsf{Top}^{\mathrm{sep}}$.

Solution 32.

1. To show that f is an epimorphism, let's consider two morphism $g, h : \mathbb{R} \to A$ with A an arbitrary separated topological space such that the following diagram

$$\begin{tikzcd} \mathbb{Q} \arrow[r, "f"] \arrow[d, "f"] & \mathbb{R} \arrow[d, "g"] \\ \mathbb{R} \arrow[r, "h"] & A \end{tikzcd}$$

commutes. For the sake of the argument let $x, y \in \mathbb{R}$ such that $g(x) \neq h(x)$ in A. Since A is separated we can consider $U \in N(g(x))$ and $V \in N(h(x))$ such that $U \cap V = \emptyset$. Since all the maps are continuous we have that $g^{-1}(U)$ and $h^{-1}(V)$ are open. Moreover, we have that $x \in g^{-1}(U) \cap h^{-1}(V)$, thus it's a non empty open set. From density of $\mathbb{Q}$ in $\mathbb{R}$ we have that

$$g^{-1}(U) \cap h^{-1}(V) \cap \mathbb{Q} \neq \emptyset$$

Hence, let $q \in g^{-1}(U) \cap h^{-1}(V) \cap \mathbb{Q}$. From the commutativity of the diagram we have

$$(g \circ f)(q) = (h \circ f)(q)$$

which means that $g(q) \in U \cap V$ and we have a contradiction. For the second part of the exercise, we can consider A as the line with two π's (that is, the line with two origins built with π instead of 0). The map g and h are two maps such that $g(x) = h(x) = x$ for $x \neq \pi$ and $g(0) = \pi_1$ and $h(0) = \pi_2$. Since $\pi \notin \mathbb{Q}$ we have that $g \circ f = h \circ f$ bat $g \neq h$.

2. In this category, the continuous injective maps are monomorphisms, thus g is a monomorphism. To prove that is an epimorphism, let consider the following commutative diagram

$$\begin{tikzcd} {[0,1)} \arrow[r, "g"] \arrow[d, "g"] & U \arrow[d, "k"] \\ U \arrow[r, "j"] & A \end{tikzcd}$$

with $k, j : U \to A$ continuous maps between separated topological spaces. If $x = 1 = e^{2\pi i} \in U$ then we have

$$k(1) = k(e^{2\pi i}) = k(e^0) = k(g(0)) = j(g(0)) = j(e^0) = j(e^{2\pi i}) = j(1)$$

If $x \neq 1$, we can write as θ_x the only number in $[0, 1]$ such that the $e^{2\pi i \theta_x} = x$, thus we have

$$k(x) = k(g(\theta_x)) = j(g(\theta_x)) = j(x)$$

Thus is an epimorphism. To prove that it is not an isomorphism, we have to show that g is not an homeomorphism. For the sake of the argument, let's suppose that g is an isomorphism, thus it's inverse g^{-1} has to be continuous. Instead, we have that $g^{-1}(1) = 0$ but for any t sufficently near to 1 we have that $g^{-1}(e^{2\pi i t}) > \frac{1}{2}$. Which means that is not continuous.

Exercise 33 (Exercise 6)**.** *A necessary condition for representability in* Top.

1. Let F be a representable contravariant functor from Top to Set. Let X be a topological space and let (U_i) be an open cover of X. For every index i, denote by u_i the inclusion $U_i \hookrightarrow X$, and for every pair (i,j) denote by u_{ij} the inclusion $U_i \cap U_j \hookrightarrow U_i$. Let (s_i) be an element of $\prod_i F(U_i)$ such that for every pair (i,j) one has

$$F(u_{ij})(s_i) = F(u_{ji})(s_j) \quad \text{in } F(U_i \cap U_j).$$

Show that there exists a unique $s \in F(X)$ such that $F(u_i)(s) = s_i$ for all i.

2. Let G be the (contravariant) subfunctor of $h^{\mathbb{R}} : \mathsf{Top} \to \mathsf{Set}$ that sends a topological space X to the set $G(X)$ of *bounded* continuous maps from X to $\mathbb{R}$. Show that G is not representable.

Solution 33.

1. Since F is representable by a topological space Y (thus for any topological space T we have a bijection $F(T) \cong h^Y(T)$), we can consider the equivalent claim with h^Y instead of F. This means that

$$h^Y(u_{ij})(s_i) = h^Y(u_{ij})(s_j) \quad \text{in } h^Y(U_i \cap U_j)$$

since we have that $(s_i) \in \prod_{i\in I} h^Y(U_i) = \prod_{i \in I} \mathrm{Hom}(U_i, Y)$ the above relation means that

$$s_i \circ u_{ij} = s_j \circ u_{ij} \quad \forall i,j \in I \tag{4}$$

We want to find an $s \in h^Y(X) = \mathrm{Hom}(X,Y)$. Now, for any $x \in X$ we can choose a $j_x \in J$ such that $x \in U_{j_x}$, then we can define

$$\begin{array}{rccc} s: & X & \to & Y \\ & x & \mapsto & s_{j_x}(x) \end{array}$$

The definition does not depend by the choose of j_x since if $x \in U_i \cap U_j$, using (4), we have $s_i(x) = s_j(x)$. We have to prove that is continuous. Let A an open set in Y, since we have that

$$s^{-1}(A) = \bigcup_i (s^{-1}(A) \cap U_i) = s_i^{-1}(A)$$

From the continuity of s_i we have that s is continuous. For the unicity, let's consider a map $q: X \to Y$ such that holds (4) with q instead of s. Thus for any $x \in X$ we have that exists a $i \in I$ such that $x \in U_i$ and we have

$$q(x) = (s_i \circ u_{ii})(x) = s(x)$$

and we have the uniqueness.

2. Using the first point of the exercise, it suffices to prove that does not hold the compatibility property of the point 1. Let's consider $X = \mathbb{R}$ and choose the open cover $\{(n, n+2)\}_{n\in\mathbb{Z}}$ for X. Let also build the bounded functions $(s_n)_{n\in\mathbb{Z}} \in \prod_{n\in\mathbb{Z}} \mathrm{Hom}((n,n+2), \mathbb{R})$ defined as

$$\begin{array}{rccc} s_n: & (n,n+2) & \to & \mathbb{R} \\ & x & \mapsto & x \end{array}$$

Those functions obviusly satisfy the condition (4). Since the only function that satisfy the restriction conditions is

$$\begin{array}{rccc} s: & \mathbb{R} & \to & \mathbb{R} \\ & x & \mapsto & x \end{array}$$

and this is not bounded, we have that we can extend the function on the all domain. Thus, reversing the implication in the first point of the exercise, we have that G can not be representable.

8 Final exam

Exercise 34 (Exercise 1)**.** Let k be a field and let C be the category of *pointed vector spaces* whose objects are pairs (V, v) where V is a k-vector space and v an element of V, a morphism from (V, v) to (W, w) being a k-linear map $f\colon V \to W$ such that $f(v) = w$.

1. Show that C admits cartesian products and coproducts, which you will describe. Show that it admits an initial object and a terminal object, which you will describe.

2. Let $G\colon \mathsf{C} \to k\text{-}\mathbf{Vect}$ be the forgetful functor $(V, v) \mapsto V$ (which sends an arrow $f\colon (V, v) \to (W, w)$ in C to the linear map $f\colon V \to W$). Show that G has a left adjoint, which you will describe. Does it have a right adjoint?

Solution 34.

1. Let $A := (V, v)$ and $B := (W, w)$ two pointed vector spaces. Let's show that $A \prod B = (V \times W, (v, w))$. Let's consider the following commutative diagram

$$\begin{tikzcd}
(T,t) \arrow[drr, bend left, "g"] \arrow[dr, bend right, "f"'] & & \\
& (V \times W, (v,w)) \arrow[r, "\pi_W"] \arrow[d, "\pi_V"] & (W,w) \\
& (V,v) &
\end{tikzcd}$$

We can define the map

$$\begin{array}{rccc} p: & (T,t) & \to & (V \times W, (v,w)) \\ & (a,b) & \mapsto & (f(a), g(b)) \end{array}$$

Clearly such a map, let the diagram be commutative and it is also the only map that can do such a thing.

Let's now prove that $(V, v) \coprod (W, w) = ((V \times W)/\rho, [(v, 0)])$ where $(a, b)\rho(c, d)$ if and only if $(a, b) - (c, d) = (v, -w)$. Let's consider the following commutative diagram

$$\begin{tikzcd}
(T,t) & & \\
& ((V \times W)/\rho, [(v,0)]) & (W,w) \arrow[l, "\pi \circ i_W"] \arrow[ull, bend right, "g"'] \\
& (V,v) \arrow[u, "\pi \circ i_V"] \arrow[uul, bend left, "f"] &
\end{tikzcd}$$

Clearly we have that $(\pi \circ i_W)(w) = \pi(0,w) = [(0,w)] = [(v,0)] = \pi(v,0) = (\pi \circ i_V)(v)$. Let's define

$$\begin{array}{rccc} T: & ((V\times W)/\rho, [(v,0)]) & \to & (T,t) \\ & ([a,b]) & \mapsto & f(a)+g(b) \end{array}$$

This is a well defined map since, if $[(a,b)] = [(c,d)]$ we have that $(a,b)-(c,d) = (v,-w)$ and we have $T([a,b]) - T([c,d]) = f(a) - g(b) - f(c) + g(d) = f(a-c) - g(b-d) = f(v) - g(w) = t - t = 0$. Clearly we have that the diagram commutes. Let $H : ((V\times W)/\rho, [(v,0)]) \to (T,t)$ be another map such that the diagram commutes, thus we have

$$f(a) = (H\circ\pi\circ i_V)(a) = H([(a,0)])$$

$$g(b) = (H\circ\pi\circ i_W)(b) = H([(0,b)])$$

Thus, for any $([a,b]) \in ((V\times W)/\rho, [(v,0)])$ we get

$$\begin{aligned} H([(a,b)]) &= H([(a,0)] + [(0,b)]) \\ &= H([(a,0)]) + H([(0,b]) \\ &= f(a) + g(b) = T([(a,b)]) \end{aligned}$$

Thus, it is unique. The final object is clearly $(\{0\},0)$. The initial object is $(k,1)$ and for any (V,v) vector the space, the unique map is

$$\begin{array}{rccc} u: & (k,1) & \to & (V,v) \\ & a & \mapsto & av \end{array}$$

2. We want to build a functor $H : k-\mathbf{Vec} \to \mathcal{C}$ such that for any V vector space and (W,w) pointed vector space we get

$$\mathrm{Hom}_{\mathcal{C}}(H(V),(W,w)) \cong \mathrm{Hom}_{k-\mathbf{Vec}}(V,W)$$

We can define

$$\begin{array}{rccc} H: & \mathbf{Vec} & \to & \mathcal{C} \\ & V & \mapsto & (V\times k,(0,1)) \\ & (f:V\to W) & \mapsto & \begin{pmatrix} (V\times k,(0,1)) & \to & (W\times k,(0,1)) \\ (v,a) & \mapsto & (f(v),a) \end{pmatrix} \end{array}$$

We can define the following maps

$$\begin{array}{rccc} u: & \mathrm{Hom}_{\mathcal{C}}((V\times k,(0,1)),(W,w)) & \to & \mathrm{Hom}_{k-\mathbf{Vec}}(V,W) \\ & f & \mapsto & f(\cdot,0) \end{array}$$

$$\begin{array}{rccc} u': & \mathrm{Hom}_{k-\mathbf{Vec}}(V,W) & \to & \mathrm{Hom}_{\mathcal{C}}((V\times k,(0,1)),(W,w)) \\ & g & \mapsto & \begin{pmatrix} (V\times k,(0,1)) & \to & (W,w) \\ (v,a) & \mapsto & (g(v)+aw) \end{pmatrix} \end{array}$$

Thus we have

$$u(u'(f)) = f, \quad u'(u(f)) = f$$

hence, H is a left adjoint for G. For the sake of the argument, let's suppose that there exists a right adjoint for G, thus, G is a left adjoint. Thus we have that

$$G(\operatorname{colim}(D)) \cong \operatorname{colim} G(D)$$

If we consider $D = \emptyset$ we get

$$k = G(k, 1) = G(\operatorname{colim}_{\mathcal{C}}(D)) \cong \operatorname{colim}_{k-\mathbb{V}} G(D) = \operatorname{colim}_{k-\mathbf{Vec}}(D) = \{0\}$$

that is not possible.

Exercise 35 (Exercise 2)**.** We recall the following definition, encountered in previous exercises: a morphism $f\colon X \to Y$ in a category C is called a *monomorphism*, resp. *epimorphism*, if for every object Z of C the map

$$\varphi \mapsto f \circ \varphi \ \text{ from } \operatorname{Hom}(Z, X) \text{ to } \operatorname{Hom}(Z, Y),$$
$$\text{resp. } h \mapsto h \circ f \ \text{ from } \operatorname{Hom}(Y, Z) \text{ to } \operatorname{Hom}(X, Z),$$

is injective.

1. Carefully prove that in **Ens** a morphism is a monomorphism, resp. an epimorphism, if and only if it is injective, resp. surjective.

2. Let E be a partially ordered set, viewed as a category ($\operatorname{Hom}(i, j)$ is a singleton if $i \leq j$ and empty otherwise). What are the monomorphisms and epimorphisms of E?

3. Let C and D be two categories and let $F\colon \mathsf{C} \to \mathsf{D}$ and $G\colon \mathsf{D} \to \mathsf{C}$ be two covariant functors such that (F, G) is an adjoint pair.

   (3a) Show that if $f\colon X \to Y$ is an epimorphism in C, then $F(f)$ is an epimorphism in D.

   (3b) Show that if $g\colon U \to V$ is a monomorphism in D, then $G(g)$ is a monomorphism in C.

4. Let C be a category. An object Z of C is called *projective* if for every epimorphism $f\colon X \to Y$ in C the map $\psi \mapsto f \circ \psi$ from $\operatorname{Hom}(Z, X)$ to $\operatorname{Hom}(Z, Y)$ is surjective.

   (4a) Show that in **Ens** every object is projective.

   (4b) Show that $\mathbb{Z}$ is a projective object of **Gp** and that $\mathbb{Z}/2\mathbb{Z}$ is not.

   (4c) Let k be a field. Show that $k[X_1, \ldots, X_n]$ is a projective object of k-**alg**.

   (4d) Let **Comp** be the full subcategory of **Top** consisting of compact spaces. Show that $[0, 1]$ is not a projective object of **Comp** (this question is difficult).

   (4e) Let F be a covariant functor from C to a category D. Suppose that F admits a right adjoint G such that $G(g)$ is an epimorphism for every epimorphism g in D. Show that if Z is a projective object of C, then $F(Z)$ is a projective object of D.

Solution 35.

1. See exercise 2.

2. Let $f : a \to b$ be a map in E. Let's consider now two maps $g, h : x \to a$ such that $f \circ g = f \circ h$. Since $\mathrm{Hom}(x, a)$ it's not grater then a singleton and it is not empty (since it contains g and h) we get that $g = h$. Thus we have that each morphism is a monomorphism. Quite the same proof holds for the epimorphism case.

3. For any X object in $\mathcal{C}$ and Z object in $\mathcal{D}$ we have the following natural isomorphism

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Z) \overset{u}{\cong} \mathrm{Hom}_{\mathcal{C}}(X, G(Z))$$

Fix also $v := u^{-1}$.

(3a) Let $f : X \to Y$ be an epimorphism in $\mathcal{C}$. Let's consider now $g, h : F(Y) \to Z$ two morphisms in $\mathcal{D}$ such that $g \circ F(f) = h \circ F(f)$. Using the above isomoprhism there exist morphisms $\alpha, \beta : Y \to G(Z)$ with $v(\alpha) = g$, $v(\beta) = h$. Using the naturality of the isomorphism, i.e. the commutativity of the diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{D}}(F(Y), Z) \arrow[r, "u"] \arrow[d, "-\circ F(f)"'] & \mathrm{Hom}_{\mathcal{C}}(Y, G(Z)) \arrow[d, "-\circ f"] \\
\mathrm{Hom}_{\mathcal{D}}(F(X), Z) \arrow[r, "u"'] & \mathrm{Hom}_{\mathcal{C}}(X, G(Z))
\end{tikzcd}$$

we get

$$\begin{aligned}
v(\alpha \circ f) &= v(\alpha) \circ F(f) \\
&= g \circ F(f) \\
&= h \circ F(f) \\
&= v(\beta) \circ F(f) \\
&= v(\beta \circ f)
\end{aligned}$$

Since v is bijection, we get

$$\alpha \circ f = \beta \circ f$$

Since f is an epimorphism, we have that $\alpha = \beta$, hence $g = h$. Thus F preserve the epimorphisms.

(3b) Let $g : U \to V$ be a monomorphism in $\mathcal{D}$. Let's consider now $f, h : Z \to G(U)$ two morphisms in $\mathcal{C}$ such that $G(g) \circ f = G(g) \circ h$. Using the above isomorphism, we get two morphisms $\alpha, \beta : F(Z) \to U$ such with $u(\alpha) = f$ and $u(\beta) = h$. Using the naturality of the foollowing diagam

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{D}}(F(Z), U) \arrow[r, "u"] \arrow[d, "g\circ -"] & \mathrm{Hom}_{\mathcal{C}}(Z, G(U)) \arrow[d, "G(g)\circ -"] \\
\mathrm{Hom}_{\mathcal{D}}(F(Z), V) \arrow[r, "u"] & \mathrm{Hom}_{\mathcal{C}}(Z, G(V))
\end{tikzcd}$$

we get

$$\begin{aligned}u(g \circ \alpha) &= (G(g) \circ -)(u(\alpha)) \\ &= G(g) \circ u(\alpha) \\ &= G(g) \circ f \\ &= G(g) \circ h \\ &= G(g) \circ u(\beta) \\ &= (G(g) \circ -)(u(\beta)) = u(g \circ \beta)\end{aligned}$$

Since u is bijection, we get

$$g \circ \alpha = g \circ \beta$$

Since g is a monomorphism, we get $\alpha = \beta$ which means $f = h$. Thus $G(g)$ is a monomoprhism, and G preserves monomorphisms.

4.

(4a) Let Z be a set and $f : X \to Y$ an epimorphism of arbitrary sets X and Y. We want to prove that the map

$$\begin{array}{ccc}\mathrm{Hom}(Z, X) & \to & \mathrm{Hom}(Z, Y) \\ \psi & \mapsto & f \circ \psi\end{array}$$

is surjective. Since f is an epimorphism, from a previous exercise, we have that f is surjective, thus for any $y \in Y$ there exists $x_y \in X$ [6] such that $f(x_y) = y$. Let's consider $g \in \mathrm{Hom}(Z, Y)$; we can define a map ψ as follows

$$\begin{array}{cccc}\psi : & Z & \to & X \\ & z & \mapsto & x_{g(z)}\end{array}$$

Clearly, for any $z \in Z$, we have that

$$(f \circ \psi)(z) = f(\psi(z)) = f(x_{g(z)}) = g(z)$$

Thus the map on Hom is surjective.

(4b) See 2 to see that an epimorphism is a surjective map in Grp.
Let's consider $\mathbb{Z}$ and let $f : X \to Y$ be an epimorphism of arbitrary groups X and Y. We want to prove that the homomorphism

$$\begin{array}{ccc}\mathrm{Hom}(\mathbb{Z}, X) & \to & \mathrm{Hom}(\mathbb{Z}, Y) \\ \psi & \mapsto & f \circ \psi\end{array}$$

is surjective. Since f is an epimorphism, from a previous exercise, we have that f is surjective, thus for any $y \in Y$ there exists $x_y \in X$ [7] such that $f(x_y) = y$. Let's consider $g \in \mathrm{Hom}(\mathbb{Z}, Y)$; we can define a map ψ as follows

$$\begin{array}{cccc}\psi : & \mathbb{Z} & \to & X \\ & n & \mapsto & n x_{g(1)}\end{array}$$

[6] We can choose this element using the axiom of choice.
[7] We can choose this element using the axiom of choice.

Clearly, for any $n \in \mathbb{Z}$, we have that

$$(f \circ \psi)(n) = f(\psi(n)) = nf(x_{g(1)}) = ng(1) = g(n)$$

We have to prove that it is an homomorphism, that is:

$$\psi(n+k) = (n+k)x_{g(1)} = nx_{g(1)} + kx_{g(1)} = \psi(n) + \psi(k)$$

Now let's prove that $\mathbb{Z}/2\mathbb{Z}$ it's not projective. Let's consider the surjective map $f : \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ sending $[1]_4 \mapsto [1]_2$ and

$$\begin{array}{ccc} \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}) & \to & \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \\ \psi & \mapsto & f \circ \psi \end{array}$$

Clearly the map

$$\begin{array}{cccc} g: & \mathbb{Z}/2\mathbb{Z} & \to & \mathbb{Z}/2\mathbb{Z} \\ & [1]_4 & \mapsto & [1]_2 \end{array}$$

is a map in $\mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$. Moreover $g \notin \mathrm{Im}(f \circ \cdot)$, indeed the only map in $\mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z})$ are 0, the zero map, and the map $h : [1]_2 \mapsto [2]_4$. Since $f \circ 0 = 0$ and $f \circ h = 0$ we get the assertion. Thus $\mathbb{Z}/2\mathbb{Z}$ it's not a projective object.

(4c) THE EXERCISE IS FALSE

(4d) Let's consider Z the disjoint union of $X = [0, 1/2]$ and $Y = [1/2, 1]$ and let $p : Z \to [0, 1]$ be the obvious continuous map sending $i \mapsto i$. For the sake of the argument, let's assume that it is projective. Thus, if we consider the map p as before, and the map $\mathrm{Id} \in \mathrm{Hom}([0, 1], [0, 1])$ we have that should exists a map $h \in \mathrm{Hom}([0, 1], Z)$ such that

$$p \circ h = \mathrm{Id}$$

Since the relation $p^{-1}(s) = \{0\}$ is only true with $s = 0$ and also $\mathrm{Id}^{-1}(0) = 0$ we get that $h(0) = 0$. In the same way, we get that $h(1) = 1$. Hence, we get that $h^{-1}(X) \neq \emptyset \neq h^{-1}(Y)$, which means that they are two closed sets non-empty. Since $\{X, Y\}$ is a covering for Z, then we have that $\{h^{-1}(X), h^{-1}(Y)\}$ is covering for $[0, 1]$. Since we have that $h^{-1}(X) \cap h^{-1}(Y) = \emptyset$ we have a contradiction with the connection of $[0, 1]$. Thus it is not projective.

(4e) In order to prove that $F(Z)$ is a projective object, we have to prove that for any epimorphism $f : A \to B$ the map

$$\begin{array}{ccc} \mathrm{Hom}(F(Z), A) & \to & \mathrm{Hom}(F(Z), B) \\ \psi & \mapsto & f \circ \psi \end{array}$$

is surjective. Using the natural bijection given by the adjunction, we get that the following diagram

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{D}}(F(Z), A) \arrow[r, "\overset{u}{\simeq}"] \arrow[d, "f\circ -"'] & \mathrm{Hom}_{\mathcal{C}}(Z, G(A)) \arrow[d, "G(f)\circ -"] \\ \mathrm{Hom}_{\mathcal{D}}(F(Z), B) \arrow[r, "\underset{v}{\simeq}"'] & \mathrm{Hom}_{\mathcal{C}}(Z, G(B)) \end{tikzcd}$$

commutes. Thus we get that

$$(f \circ -) = v^{-1} \circ (G(f) \circ -) \circ u$$

That is

$$(f_*) = v^{-1} \circ G(f)_* \circ u$$

Now, let consider an element $\varphi \in \operatorname{Hom}(F(Z), B)$ and we can consider $v(\varphi) \in \operatorname{Hom}_{\mathcal{C}}(Z, G(B))$. Since Z is a projective object in $\mathcal{C}$, f is an epimorphism and G sends epimorphisms in epimorphisms, we get that there exists $h \in \operatorname{Hom}_{\mathcal{C}}(Z, G(A))$ such that

$$v(\varphi) = G(f) \circ h = G(f)_*(h)$$

Thus, if we consider the element $u^{-1}(h) \in \operatorname{Hom}_{\mathcal{C}}(F(Z), A)$, from the commutativity of the above diagram, we get that

$$(f_*)(u^{-1}(h)) = (v^{-1} \circ G(f)_* \circ u)(u^{-1}(h)) = (v^{-1} \circ G(f)_*)(h) = v^{-1}(v(\varphi)) = \varphi$$

Thus, the map is surjective.

Exercise 36 (Exercise 3)**.** Recall that `Ann` denotes the category of unital commutative rings.

1. Show that the covariant forgetful functor F from `Ann` to `Set` is representable, and give a representing object.

2. Describe the set of endomorphisms of F.

3. Let n be an integer. Show that the functor T_n from `Ann` to `Set` sending A to the set of elements a of A such that $na = 0$ is representable, and give a representing object.

4. Describe the set of endomorphisms of T_n (this question is delicate).

Solution 36.

1. Let's prove that $\mathbb{Z}[x]$ is a representative for F. In order to do this, let's show that

$$h_{\mathbb{Z}[x]} \cong F$$

that is equivalent to show the bijection

$$h_{\mathbb{Z}[x]}(A) \cong F(A) = A$$

for any ring A. Let's consider the maps

$$\begin{array}{rccc} f: & \operatorname{Hom}(\mathbb{Z}[x], A) & \to & A \\ & \psi & \mapsto & \psi(x) \end{array}$$

and

$$\begin{array}{rccc} g: & A & \to & \operatorname{Hom}(\mathbb{Z}[x], A) \\ & a & \mapsto & \begin{pmatrix} \psi_a: & \mathbb{Z}[x] & \to & A \\ & x & \mapsto & a \end{pmatrix} \end{array}$$

Now, for any $a \in A$ we have

$$(f \circ g)(a) = f(\psi_a) = \psi_a(x) = a$$

For any map $\psi \in \mathrm{Hom}(\mathbb{Z}[x], A)$ we have

$$(g \circ f)(\psi) = g(\psi(x)) = \psi_{\psi(x)}$$

and we have that

$$\psi_{\psi(x)}(x) = \psi(x)$$

thus it's bijective. Then a representative for H is $(\mathbb{Z}[x], x)$.

2. In general, for two representable morphisms H, G with representatives (X, ξ) and (Y, η), we have that the following map

$$\begin{array}{ccc} \mathrm{Hom}(Y, X) & \to & \mathrm{End}(H, G) \\ f & \mapsto & u_{G,\eta} \circ f^* \circ u_{H,\xi}^{-1} \end{array}$$

is an isomorphism with inverse the map such that sends a natural transformation ψ to the unique map $f : Y \to X$ such that $\psi(X)(\xi) = G(f)(\eta)$. Let's use the previous formula in the case $H = G = F$, $X = Y = \mathbb{Z}[x]$ and $\xi = \eta = x$. We get

$$\begin{array}{ccc} \mathrm{Hom}(\mathbb{Z}[x], \mathbb{Z}[x]) & \to & \mathrm{End}(F, F) \\ f & \mapsto & u_{F,x} \circ f^* \circ u_{F,\xi}^{-1} \end{array}$$

Thus, each endomorphism is of the form $u_{F,x} \circ f^* \circ u_{F,\xi}^{-1}$. Let's be more precise: let R be a ring, then we have

$$\begin{array}{rccc} u_{F,\xi}^{-1}(R): & F(R) = R & \to & h_{\mathbb{Z}[x]}(R) = \mathrm{Hom}(\mathbb{Z}[x], R) \\ & r & \mapsto & \begin{pmatrix} \phi_r : & \mathbb{Z}[x] & \to & R \\ & x & \mapsto & r \end{pmatrix} \end{array}$$

$$\begin{array}{rccc} f^*(R): & h_{\mathbb{Z}[x]}(R) = \mathrm{Hom}(\mathbb{Z}[x], R) & \to & h_{\mathbb{Z}[x]}(R) = \mathrm{Hom}(\mathbb{Z}[x], R) \\ & \phi & \mapsto & \phi \circ f \end{array}$$

$$\begin{array}{rccc} u_{F,\xi}(R): & h_{\mathbb{Z}[x]}(R) = \mathrm{Hom}(\mathbb{Z}[x], R) & \to & F(R) = R \\ & \phi & \mapsto & \phi(x) \end{array}$$

thus for a ring R we have

$$\begin{array}{rccc} u_{F,x} \circ f^* \circ u_{F,\xi}^{-1}(R): & F(R) = R & \to & F(R) = R \\ & r & \mapsto & (\phi_r \circ f)(x) \end{array}$$

Moreover we have that an endomorphism of $\mathbb{Z}[x]$ is such that $x \mapsto p(x)$ for a polonomial, thus we can say that each morphism f is of the form f_p where

$$\begin{array}{rccc} f_p : & \mathbb{Z}[x] & \to & \mathbb{Z}[x] \\ & x & \mapsto & p(x) \end{array}$$

In this way we have that an arbitrary endomorphism for F is of the form

$$\begin{array}{rrcl} u_{F,x} \circ f^* \circ u_{F,\xi}^{-1}(R): & F(R) = R & \to & F(R) = R \\ & r & \mapsto & (\phi_r \circ f_p)(x) \end{array}$$

Since $(\phi_r \circ f_p)(x) = \phi_r(p(x)) = p(r)$ we get

$$\begin{array}{rrcl} u_{F,x} \circ f_p^* \circ u_{F,\xi}^{-1}(R): & F(R) = R & \to & F(R) = R \\ & r & \mapsto & p(r) \end{array}$$

3. Let's prove that $\mathbb{Z}[x]/(nx)$ is a representative for T_n. In order to do this, let's show that

$$h^{\mathbb{Z}[x]/(nx)} \cong T_n$$

that is equivalent to show the bijection

$$h^{\mathbb{Z}[x]/(nx)}(A) \cong T_n(A)$$

for any ring A. Let's consider the maps

$$\begin{array}{rccl} f: & \mathrm{Hom}(\mathbb{Z}[x]/(nx), A) & \to & T_n(A) \\ & \psi & \mapsto & \psi(\bar{x}) \end{array}$$

and

$$\begin{array}{rccc} g: & T_n(A) & \to & \mathrm{Hom}(\mathbb{Z}[x]/(nx), A) \\ & a & \mapsto & \left(\begin{array}{rccc} \psi_a: & \mathbb{Z}[x]/(nx) & \to & A \\ & \bar{x} & \mapsto & a \end{array}\right) \end{array}$$

Both maps are well defined since for f:

$$nf(\psi) = n\psi(\bar{x}) = \psi(n\bar{x}) = \psi(\bar{0}) = 0$$

and for g: ψ_a is well defined as a map out of $\mathbb{Z}[x]/(nx)$ since the only relation we need to respect is $nx = 0$, and indeed $n\psi_a(\bar{x}) = na = 0$ since $a \in T_n(A)$.

Now, for any $a \in T_n(A)$ we have

$$(f \circ g)(a) = f(\psi_a) = \psi_a(\bar{x}) = a$$

For any map $\psi \in \mathrm{Hom}(\mathbb{Z}[x]/(nx), A)$ we have

$$(g \circ f)(\psi) = g(\psi(\bar{x})) = \psi_{\psi(\bar{x})}$$

and since a ring homomorphism out of $\mathbb{Z}[x]/(nx)$ is uniquely determined by the image of $\bar{x}$, and $\psi_{\psi(\bar{x})}(\bar{x}) = \psi(\bar{x})$, we conclude $\psi_{\psi(\bar{x})} = \psi$. Thus the maps are mutually inverse bijections, and a representative for T_n is $(\mathbb{Z}[x]/(nx),\ \bar{x})$.

4. Let's use the formula from the second point of this exercise in the case $H = G = T_n$, $X = Y = \mathbb{Z}[x]/(nx)$ and $\xi = \eta = \bar{x}$. We get

$$\begin{array}{rcc} \mathrm{Hom}(\mathbb{Z}[x]/(nx), \mathbb{Z}[x]/(nx)) & \to & \mathrm{End}(T_n, T_n) \\ f & \mapsto & u_{T_n,\bar{x}} \circ f^* \circ u_{T_n,\bar{x}}^{-1} \end{array}$$

Thus, each endomorphism is of the form $u_{T_n,\bar{x}} \circ f^* \circ u_{T_n,\bar{x}}^{-1}$. Let's be more precise: let R be a ring, then we have

$$u_{T_n,\bar{x}}^{-1}(R): \begin{array}{ccc} T_n(R) & \to & h_{\mathbb{Z}[x]/(nx)}(R) = \mathrm{Hom}(\mathbb{Z}[x]/(nx), R) \\ r & \mapsto & \begin{pmatrix} \phi_r: & \mathbb{Z}[x]/(nx) & \to & R \\ & \bar{x} & \mapsto & r \end{pmatrix} \end{array}$$

$$f^*(R): \begin{array}{ccc} h_{\mathbb{Z}[x]/(nx)}(R) = \mathrm{Hom}(\mathbb{Z}[x]/(nx), R) & \to & h_{\mathbb{Z}[x]/(nx)}(R) = \mathrm{Hom}(\mathbb{Z}[x]/(nx), R) \\ \phi & \mapsto & \phi \circ f \end{array}$$

$$u_{T_n,\bar{x}}(R): \begin{array}{ccc} h_{\mathbb{Z}[x]/(nx)}(R) = \mathrm{Hom}(\mathbb{Z}[x]/(nx), R) & \to & T_n(R) \\ \phi & \mapsto & \phi(\bar{x}) \end{array}$$

Thus for a ring R we have

$$u_{T_n,\bar{x}} \circ f^* \circ u_{T_n,\bar{x}}^{-1}(R): \begin{array}{ccc} T_n(R) & \to & T_n(R) \\ r & \mapsto & (\phi_r \circ f)(\bar{x}) \end{array}$$

Moreover, an endomorphism of $\mathbb{Z}[x]/(nx)$ is determined by $\bar{x} \mapsto p(\bar{x})$ for a polynomial of the form

$$p(x) = \sum_{i \geq 1} a_i x^i, \quad a_i \in \mathbb{Z}$$

where the constant term is zero since $p(\bar{x})$ must be of n-torsion in $\mathbb{Z}[x]/(nx)$. Thus each morphism f is of the form f_p where

$$f_p: \begin{array}{ccc} \mathbb{Z}[x]/(nx) & \to & \mathbb{Z}[x]/(nx) \\ \bar{x} & \mapsto & p(\bar{x}) \end{array}$$

Moreover, since $(\phi_r \circ f_p)(\bar{x}) = \phi_r(p(\bar{x})) = p(r)$ we get

$$u_{T_n,\bar{x}} \circ f_p^* \circ u_{T_n,\bar{x}}^{-1}(R): \begin{array}{ccc} T_n(R) & \to & T_n(R) \\ r & \mapsto & p(r) \end{array}$$

Thus $\mathrm{Nat}(T_n, T_n)$ is in bijection with $\left\{\sum_{i\geq 1} a_i x^i,\ a_i \in \mathbb{Z}\right\}$, where each polynomial $p(x)$ corresponds to the natural transformation $r \mapsto p(r)$.